

A report for end semester (7th) evaluation of
B.Tech Project (BCS-753) Entitled
Space Weather Prediction Platform (CosmoPredict)
Project Report to be submitted in partial fulfillment of the requirements
for the award of the degree of
BACHELOR OF TECHNOLOGY
in
ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

By Project Group E-03

Divyansh Srivastava (2201201640028)

Krishna Kant Sharma (2201201640035)

Ajay Kumar (2201201640008)

Shantanu Shahi (2201201640055)

Under the Guidance of

Dr. SK Pandey (Professor)



Department of Computer Science & Engineering
INSTITUTE OF TECHNOLOGY & MANAGEMENT GORAKHPUR
(AFFILIATED TO DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW)
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CANDIDATE’S STATEMENT OF ORIGINALITY

We, the undersigned candidates, hereby declare that the project titled:

“Space Weather Prediction Platform (CosmoPredict)”

is an **original piece of work** carried out by our project group as part of the requirements for the completion of our academic curriculum.

We affirm that:

1. **The work presented in this project is solely the result of our own effort**, investigation, and research.
2. **No part of this project has been submitted previously** to any university, institution, or publication for the award of any degree, diploma, or certificate.
3. **All sources of information used have been duly acknowledged**, and references have been clearly cited wherever applicable.
4. The datasets, models, designs, and implementation used in this project are **developed or adapted by the team**, with proper credit given for publicly available resources.
5. The project has been completed under the **guidance of our Project Guide**, and all ethical and academic standards have been strictly followed.

We fully understand that **any violation of academic integrity** or plagiarism may lead to disciplinary actions as per institutional norms.

We hereby submit this project with full responsibility for the authenticity and originality of the content.

STUDENT DECLARATION

Name of Student	Roll No.	Signature
Divyansh Srivastava	2201201640028	_____
Krishna Kant Sharma	2201201640035	_____
Ajay Kumar	2201201640008	_____
Shantanu Shahi	2201201640055	_____

PROJECT GUIDE CERTIFICATION

I hereby certify that the above candidates have carried out the project work titled

“Space Weather Prediction Platform (CosmoPredict)”

under my supervision and guidance.

To the best of my knowledge, the work reported in this project is original and has not been submitted elsewhere.

Project Guide Name: Dr. SK Pandey

Designation: Professor

Signature: _____

Date: _____

HEAD OF DEPARTMENT (HOD) APPROVAL

This is to certify that the project entitled

“Space Weather Prediction Platform (CosmoPredict)”

submitted by the above candidates has been reviewed and approved for submission as per the academic guidelines.

Head of Department (HOD): Dr. Ashutosh Kumar Rao

Department: Computer Science & Engineering

Signature: _____

Date: _____

ACKNOWLEDGEMENT

We take this opportunity to express our sincere gratitude to everyone who has supported us during the initial development and ongoing progress of our project, “**Space Weather Prediction Platform (CosmoPredict).**”

We extend our heartfelt thanks to our **Project Guide, Dr. SK Pandey**, for his continuous guidance, valuable suggestions, and motivation throughout the early phases of this work. Their expertise has been instrumental in helping us shape the foundation of the project.

We are also grateful to the **Head of Department, Dr. Ashutosh Kumar Rao**, for providing us with the opportunity, resources, and academic environment to initiate this project and carry it forward effectively. Our appreciation goes to all the **faculty members of the Department of Computer Science & Engineering**, whose consistent encouragement and constructive feedback have supported us in enhancing the quality of our work.

We would also like to acknowledge the contributions of scientific organizations such as **NASA, NOAA, and ACE**, whose publicly accessible datasets and research materials are helping us build the core predictive components of CosmoPredict.

We thank our families and friends for their understanding, patience, and constant support during the development journey.

Finally, we express our gratitude to each member of our project team for their dedication, collaboration, and commitment. Although the project is still in progress, we remain motivated to complete **CosmoPredict** with accuracy, innovation, and excellence.

— **Project Group E-03**

Divyansh Srivastava (2201201640028)

Krishna Kant Sharma (2201201640035)

Ajay Kumar (2201201640008)

Shantanu Shahi (2201201640055)

ABSTRACT

The increasing dependence on satellite-based systems for navigation, communication, defense, and space exploration has amplified the need for accurate and timely prediction of space weather events. Solar flares, coronal mass ejections (CMEs), and geomagnetic storms pose significant risks to satellites, astronauts, power grids, and global communication networks. To address these challenges, this project proposes **CosmoPredict — a Machine Learning-driven Space Weather Prediction Platform** designed to forecast critical space weather phenomena and provide actionable insights for operational safety.

CosmoPredict integrates historical and real-time data from NASA, NOAA, and ACE, processes it through advanced deep learning models such as LSTM and GRU, and generates predictive outputs for solar activity and geomagnetic disturbances. The platform includes a visualization dashboard built using Django, offering real-time analytics, risk-level indicators, and alert systems for early warnings. Additionally, a Route Diversion Module is incorporated to suggest safe orbital adjustments for satellites during high-risk events.

By combining artificial intelligence, time-series modeling, and space science, this project aims to enhance situational awareness, improve satellite resilience, and support future space missions. Although the platform is currently under development, it demonstrates strong potential to contribute to research, safety planning, and sustainable space operations.

CHAPTER 1

MOTIVATION & BACKGROUND

1.1 Introduction

In today's technologically interconnected world, space-based infrastructure has become a fundamental backbone of global systems. From aviation and maritime navigation to telecommunications, weather forecasting, financial transactions, and national defense, satellites enable seamless functioning of critical operations. Every time a user accesses GPS, checks real-time weather updates, makes an online payment, or communicates through a satellite network, space-based technologies ensure uninterrupted performance.

However, as our dependence on satellites increases, a major vulnerability becomes evident—**space weather**. The dynamic activity of the Sun, including solar flares, coronal mass ejections (CMEs), and geomagnetic storms, poses significant risks to satellites, astronauts, and technological systems on Earth.

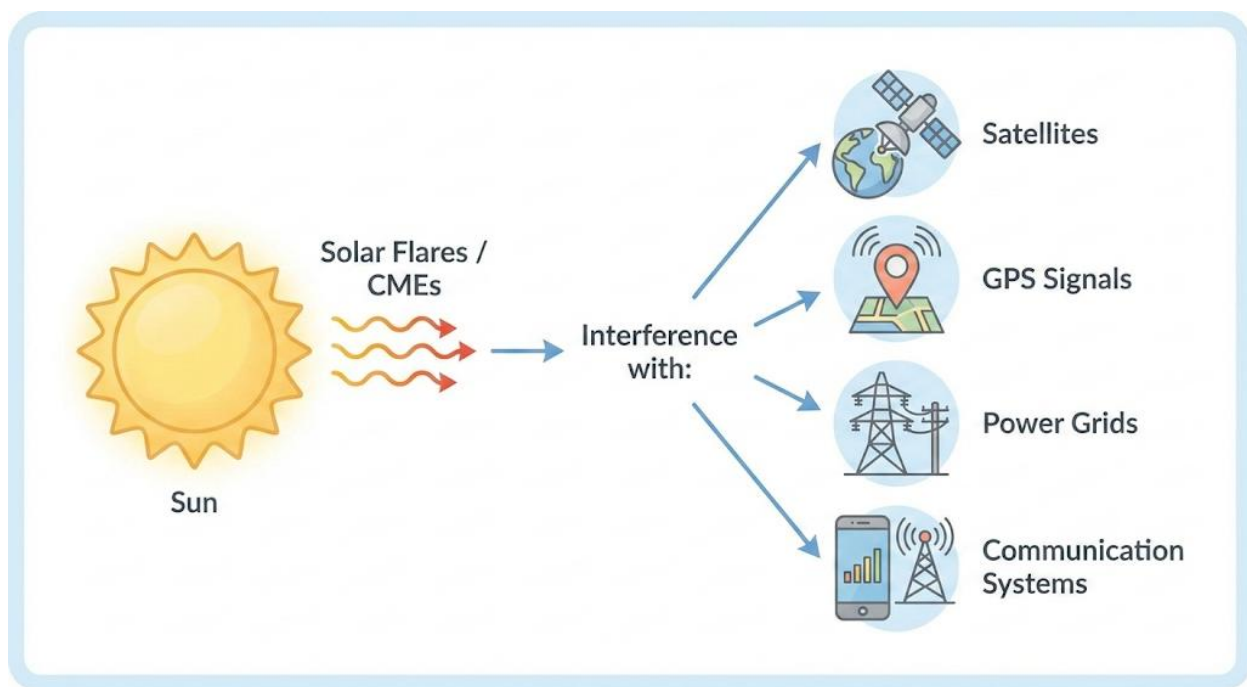


Fig 1.1: Space Weather Impact

1.2 Understanding Space Weather

Space weather refers to conditions on the Sun, in the solar wind, and within Earth's magnetosphere and ionosphere that can influence space-borne and ground-based technological systems. Key events include:

- **Solar Flares** – Sudden bursts of radiation that can disrupt radio communication.
- **Coronal Mass Ejections (CMEs)** – Massive clouds of magnetized plasma that can damage

satellites.

- **Geomagnetic Storms** – Disturbances in Earth’s magnetic field affecting navigation and power grids.

These events can occur unexpectedly and with varying intensities, making accurate prediction both challenging and essential.

1.3 Growing Dependence on Space-Based Systems

1.3.1 Expansion of Satellite Constellations

With advancements in aerospace technology, the number of operational satellites has risen dramatically.

- More than **8,000 satellites** are currently in orbit.
- By **2030**, this figure may exceed **50,000**, driven by constellations like Starlink and OneWeb.

The rapid increase magnifies the potential impact of space weather disturbances across global systems.

1.3.2 Growth in Human Space Missions

Recent developments reflect a renewed focus on manned space exploration:

- **NASA Artemis** aims to return humans to the Moon.
- **SpaceX Starship** targets interplanetary travel.
- **ISRO Gaganyaan** will send Indian astronauts to space.

These missions demand highly accurate monitoring of solar radiation events to safeguard astronaut health and mission integrity.

1.3.3 Dependence of Critical Infrastructure

Modern society relies on satellites for:

- Navigation (GPS, GLONASS, NavIC)
- Financial transactions
- Internet and broadcast communication
- Defense and surveillance

A single severe solar storm can disrupt multiple sectors simultaneously, causing global-scale economic and operational challenges.

1.4 Why This Project is Needed

Space weather prediction is complicated due to the Sun’s inherently chaotic behavior. Current models lack robust AI-driven forecasting capability and real-time operational insight. This project addresses these gaps by combining machine learning, data-driven analysis, and visualization systems.

1.4.1 Increasing Satellite Vulnerability

As satellite density grows, even small geomagnetic disturbances can lead to:

- Satellite drag
- Orbital decay
- Communication blackouts
- Sensor malfunction
- Power subsystem failures

Predictive tools become indispensable to protect expensive assets.

1.4.2 Human Safety Concerns

Astronauts face heightened risks from sudden solar energetic particle (SEP) events, which can damage biological tissues and electronic equipment.

1.4.3 Limitations of Current Forecasting Systems

Existing systems face challenges such as:

- Fragmented datasets across NASA, NOAA, ACE
- Delayed or generalized alerts
- Lack of operational recommendations for satellite safety

CosmoPredict aims to integrate these aspects into a unified, intelligent system.

1.5 Project Vision

The vision of the **Space Weather Prediction Platform (CosmoPredict)** is to create an intelligent, AI-enhanced environment capable of detecting, predicting, and responding to space weather threats.

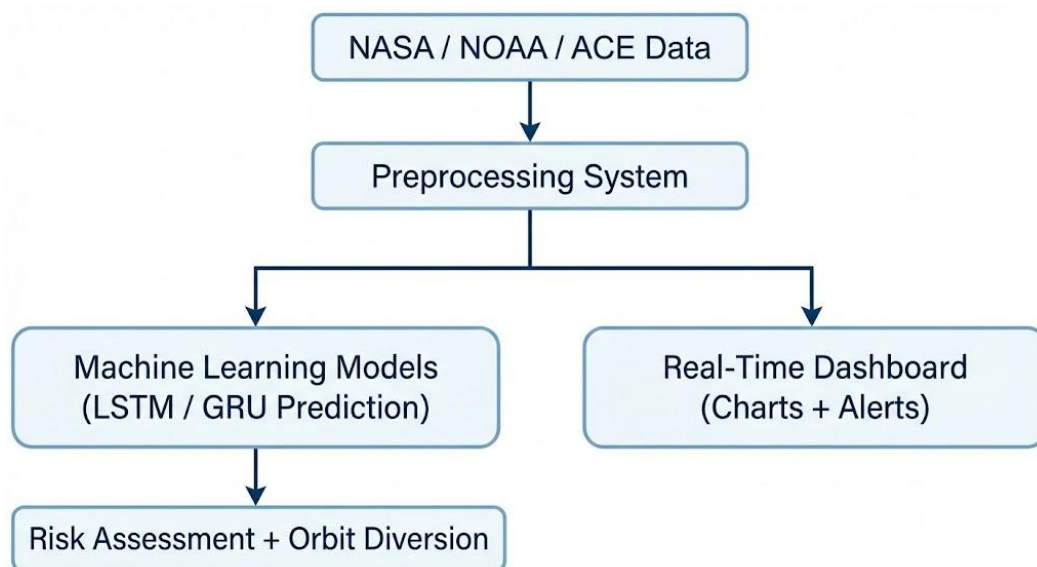


Fig 1.2: CosmoPredict Workflow Diagram

1.5.1 Core Capabilities

The proposed platform provides:

- **Real-time visualization dashboards** for monitoring solar wind, geomagnetic indices, and flare probabilities.
- **Predictive insights** using LSTM and GRU models for time-series forecasting.
- **Alert and notification systems** for at-risk satellites and missions.
- **A route diversion module** suggesting alternative orbital paths during severe disturbances.

1.6 Societal & Scientific Benefits

The CosmoPredict platform contributes meaningfully to both scientific research and societal safety.

1.6.1 Early Hazard Detection

Early alerts for solar storms allow:

- Satellite operators to switch safe modes
- Space agencies to delay EVA (spacewalks)
- Power grids to stabilize operations

1.6.2 Enhanced Astronaut Safety

Predicting solar particle events ensures:

- Better shielding preparation
- Safe scheduling of extravehicular missions
- Improved mission reliability

1.6.3 Operational Resilience for Satellites

With predictive intelligence, satellite operators can:

- Adjust orientations
- Limit power usage
- Temporarily disable vulnerable components

1.6.4 Support for Research & Policy

Researchers gain improved datasets and models for study, while policymakers can develop stronger guidelines for satellite safety and space traffic management.

1.6.5 Sustainable Space Operations

Mitigating damage prolongs satellite lifespan and reduces space debris — essential for long-term space sustainability.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

A literature survey is essential for understanding the current state of research in space weather prediction, machine learning-based forecasting systems, and satellite risk assessment techniques. The field of space weather has expanded significantly due to improvements in solar observatories, deep learning algorithms, and real-time space monitoring missions. Although considerable progress has been made, major gaps remain in prediction accuracy, data fusion, real-time implementation, and operational decision support [1], [2]. This chapter reviews prior contributions, existing forecasting systems, machine learning approaches, and highlights the research gaps motivating the development of the CosmoPredict platform.

2.2 Overview of Space Weather Research

Space weather research gained momentum when studies revealed that solar disturbances significantly impact communication systems, navigation, satellites, and power infrastructure [3], [4]. Major organizations such as NASA, NOAA, and ESA continue to contribute foundational work through space missions, data repositories, and operational alerts.

Key Findings from Previous Research

- ❖ Geomagnetic storms are strongly influenced by solar wind parameters such as velocity, proton density, and IMF [5], [6].
- ❖ Earlier prediction models relied on statistical or rule-based techniques with limited accuracy [7].
- ❖ Machine learning has shown significant improvements in forecasting precision [8], [9].
- ❖ Real-time systems exist, but integration with AI-based prediction remains limited [10].

2.3 Existing Space Weather Monitoring Systems

Several global organizations provide monitoring, visualization, and warnings for space weather events. However, most platforms do **not** provide predictive intelligence or satellite-specific operational guidance.

- ❖ NASA OMNIWeb provides historical datasets but lacks real-time forecasting [11].
- ❖ NOAA SWPC delivers alerts but relies largely on physics-based models rather than deep learning [12].
- ❖ ACE provides reliable upstream solar wind data but no predictive features [13].
- ❖ ESA SSA focuses on radiation monitoring but lacks ML-driven recommendations [14].

2.3.1 NASA OMNIWeb

- Provides long-term solar wind and geomagnetic datasets.
- Offers historical, hourly and 1-minute resolution data.
- **Strength:** Highly comprehensive datasets.
- **Limitation:** No forecasting capability [11].

2.3.2 NOAA Space Weather Prediction Center (SWPC)

- Issues global space weather alerts and warnings.
- Uses physics-based models and observational data.
- **Strength:** Reliable alert dissemination.
- **Limitation:** Alerts are broad, slow, and not satellite-specific [12].

2.3.3 ACE (Advanced Composition Explorer)

- Offers near-Earth solar wind and magnetic field data.
- **Strength:** High-frequency real-time observations.
- **Limitation:** Data gaps and no predictive modelling [13].

2.3.4 ESA Space Situational Awareness Program

- Focuses on radiation storms, flare warnings, and imaging.
- **Limitation:** Does not use ML-based forecasting or automated recommendations [14].

System	Main Functionality	Strengths	Limitations
NASA OMNIWeb	Historical space weather data	Large datasets	No real-time prediction
NOAA SWPC	Alerts, solar monitoring	Reliable global warnings	Alerts lack precision
ACE Satellite	Real-time solar wind	Minute-level updates	Data transmission gaps
ESA SSA	Radiation and flare monitoring	Detailed solar imagery	No ML forecasting

Table 2.1: Comparison of Existing Space Weather Platforms

2.4 Machine Learning in Space Weather Prediction

Recent studies highlight the growing importance of AI and deep learning in enhancing prediction accuracy. This has opened new pathways for forecasting that outperform classical statistical or physics-based models.

2.4.1 Use of LSTM Models

LSTMs capture long-term dependencies and help predict geomagnetic indices effectively.

- Camporeale improved Kp predictions using LSTM networks [1].
- Wu & Lundstedt demonstrated early neural network applications for geomagnetic forecasting [15].
- Tan et al. applied LSTM to solar flare prediction [16].

Research Example:

M. Camporeale (2019) demonstrated that LSTM models improved geomagnetic index forecasting (Kp, Dst) compared to autoregressive models.

2.4.2 Use of GRU Models

GRUs show improved training speed with competitive accuracy.

- Gruet et al. predicted solar wind speed using GRU models [17].
- Chen et al. predicted CME arrival times using GRU-based architectures [18].

Research Example:

Gruet et al. (2018) used GRU for predicting solar wind speed with promising results.

2.4.3 Hybrid Deep Learning Models

Hybrid models such as CNN-LSTM, autoencoders, and attention models improve performance:

- Bobra & Couvidat used feature extraction + SVM for flare classification [19].
- Nishizuka et al. demonstrated deep CNN-based flare prediction [20].
- Li et al. incorporated attention mechanisms for improved flare forecasting [21].

Research Insight:

Hybrid models show improved flare classification accuracy but require large datasets and computational resources.

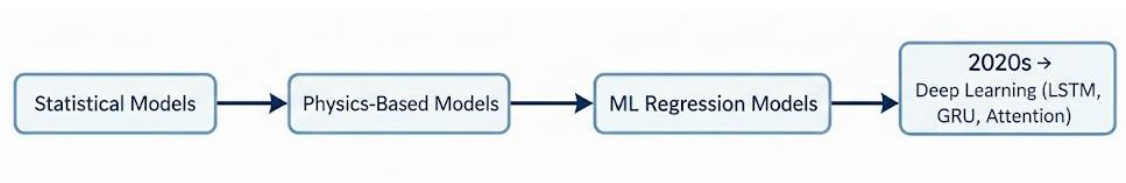


Fig 2.1: Growth of AI Approaches in Space Weather Research

2.5 Existing Gaps in Literature

Although many research studies focus on specific aspects of space weather, major gaps persist in practical applications:

2.5.1 Gap 1: Lack of Integrated Data Platforms

Most research papers use:

- A single dataset (e.g., only ACE or only OMNI)
- Limited features

- Short-term prediction windows

There is no unified system that integrates NASA + NOAA + ACE into a real-time predictive pipeline [22].

2.5.2 Gap 2: Absence of Operational Decision Support

No published research offers:

- Satellite-specific risk analysis
- Orbit diversion recommendations
- Automated decision support tools

This is a critical requirement for agencies like ISRO, NASA, and private satellite companies [14].

2.5.3 Gap 3: Limited Real-Time Implementation

Many models are trained offline and not implemented in:

- Web dashboards
- Real-time alert systems
- Live visualization tools

This restricts their practical usability [10].

2.5.4 Gap 4: Underutilization of Deep Learning

While LSTM and GRU methods are explored academically, they are rarely deployed in:

- Real-time operational systems
- User-friendly interfaces
- Integrated monitoring + prediction platforms

Advanced architectures are explored academically but not translated into operational systems [21].

CosmoPredict addresses these shortcomings.

2.6 Comparative Study of Previous Work

Research Work	Technique Used	Strength	Limitation
Camporeale (2019)	LSTM for Kp index	High accuracy	No operational interface
Gruet (2018)	GRU for solar wind	Fast training	Limited features used
Bobra (2015)	SVM for flare classification	Good classification	Weak time prediction ability
Papaioannou (2020)	Statistical CME models	Good historical insight	Poor predictive power
NOAA SWPC	Rule-based alerts	Reliable warnings	Slow + not AI-driven

Table 2.2: Analysis of Related Literature

Camporeale [1], Gruet [17], Bobra [19], Papaioannou [23], and NOAA SWPC [12] illustrate strengths but show limitations in real-time, AI-driven operational decision support.

2.7 Research Contribution of CosmoPredict

Based on the literature review, CosmoPredict contributes the following advancements:

2.7.1 Unified Data Integration

Combines multiple sources (NASA, NOAA, ACE) into a single pipeline—rarely done in existing research.

2.7.2 Real-Time AI Predictions

Implements LSTM/GRU models in a real-time environment with automated updates.

2.7.3 Actionable Satellite Risk Guidance

Introduces a **Route Diversion Module**, a feature not found in any existing academic or industrial system.

2.7.4 Interactive, User-Friendly Dashboard

Bridges research models with operational needs through:

- Charts
- Alerts
- Risk indicators

2.7.5 Practical Utility Beyond Research

Supports:

- Space missions
- Telecommunication operators
- Navigation systems
- Earth-based power grid preparation

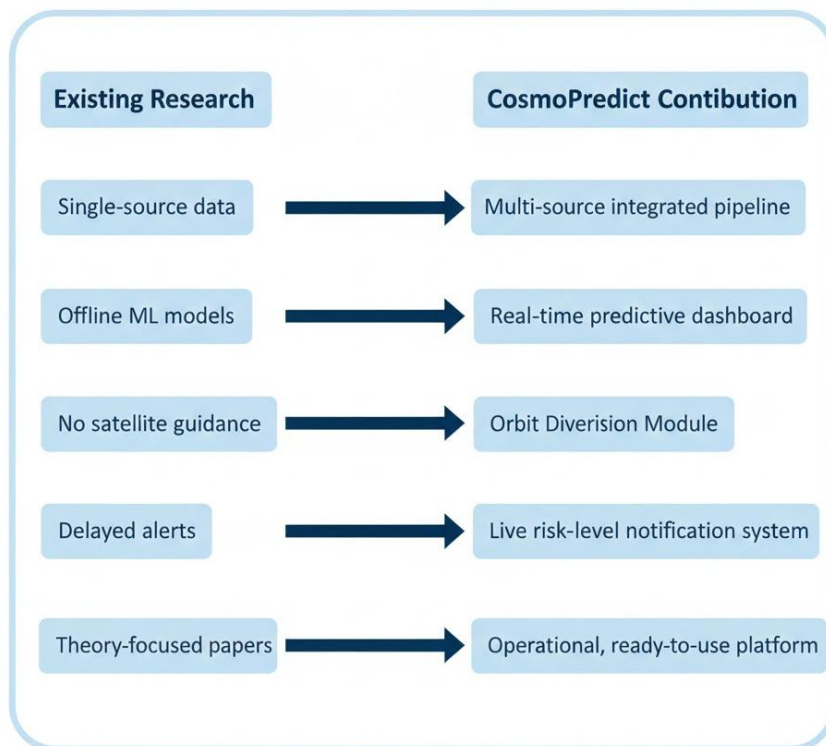


Fig 2.2: Literature Gap Mapping

CHAPTER 3

PROBLEM DEFINITION

3.1 Introduction

Despite significant advancements in astrophysics, solar physics, and atmospheric modeling, **space weather forecasting remains one of the most challenging scientific domains**. Although modern satellites and observatories continuously collect massive volumes of data, the complexity and unpredictability of solar activity make it difficult to convert this data into accurate, real-time forecasts. A major limitation of existing systems is the absence of intelligent, data-driven platforms capable of integrating diverse datasets and producing timely predictions to protect satellites, astronauts, and critical infrastructure on Earth.

This chapter defines the core problems addressed by the **Space Weather Prediction Platform (CosmoPredict)** and outlines the objectives and solutions proposed by the project.

3.2 Existing Challenges in Space Weather Prediction

Space weather forecasting faces multiple technical and operational challenges that limit the effectiveness of current systems.

3.2.1 Unpredictability of Solar Activity

Solar flares, CMEs, and high-energy particle events can occur suddenly without clear precursors. Their intensity and direction are highly variable, making physics-based prediction difficult.

Key Issues

- Sudden events with little advance warning
- Highly dynamic magnetic field interactions
- Difficulty in estimating Earth impact probability

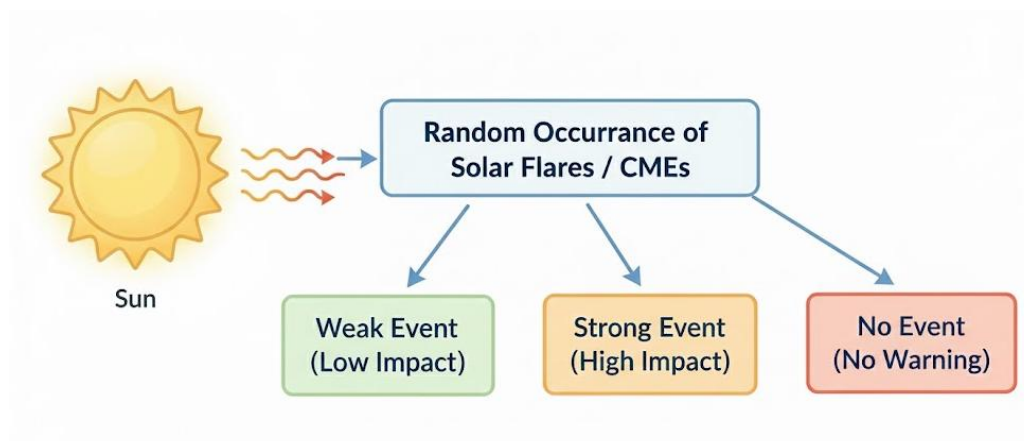


Fig 3.1: Unpredictable Nature of Solar Events

3.2.2 Fragmented Data Sources

Space weather data is distributed across multiple agencies such as:

- NASA OMNIWeb
- NOAA SWPC
- ACE satellite missions

However, these datasets differ in:

- Format
- Frequency of updates
- Availability
- Measurement techniques

This creates operational challenges for researchers and satellite operators.

3.2.3 Limited Early Warning Systems

Current space weather alerts are often:

- Delayed and Generalized
- Not specific to individual satellite missions
- Unable to provide precise risk assessment

For mission-critical operations, even a delay of minutes can result in irreversible damage.

3.2.4 Lack of Operational Guidance for Satellite Operators

Existing systems only *monitor* and *report* space weather—they do not provide **actionable recommendations**.

Missing Functionalities

- Suggested alternative orbits
- Calculated safe-zone trajectories
- Mission-specific shielding advice
- Automated risk scoring

This leaves operators without clear mitigation strategies during solar storms.

3.3 Summary of Challenges

Challenge Area	Description
Unpredictability of Solar Events	Sudden solar flares and CMEs occur without warning
Fragmented Data Sources	No unified platform integrating NASA, NOAA, and ACE data
Limited Early Alerts	Alerts are delayed or insufficiently precise
Lack of Operational Guidance	No system suggests protective satellite maneuvers or route changes

Table 3.1: Overview of Key Challenges in Space Weather Forecasting

3.4 Project Objectives

To address the limitations identified, the **CosmoPredict** system is designed with the following objectives:

3.4.1 Develop ML-Based Predictive Models

Use LSTM and GRU architectures to predict:

- Solar flare intensity
- CME likelihood
- Geomagnetic storm severity

3.4.2 Create a Real-Time Visualization Platform

A Django-based dashboard will:

- Display live solar wind parameters
- Show predicted space weather conditions
- Visualize risk levels through dynamic charts

3.4.3 Implement a Multi-Level Alert System

Provide notifications via:

- Web alerts
- Email
- SMS
- Color-coded risk indicators (green, yellow, red)

3.4.4 Introduce a Route Diversion Module

An intelligent system that:

- Analyzes satellite vulnerability
- Suggests safer orbital paths
- Provides downloadable TLE formats

3.4.5 Ensure High Model Accuracy

Validation using:

- RMSE
- MAE
- F1 Score
- Historical event comparison

3.5 Mapping Problems to Solutions

Problem Area	Proposed Solution
Lack of Predictive Intelligence	LSTM/GRU models trained on decades of space weather data
Fragmented Data	Centralized pipeline integrating NASA, NOAA, ACE APIs
No Real-Time Alerts	Automated multi-channel alert system with risk indicators
No Operational Support	Route Diversion Module generating safe orbital recommendations

Table 3.2: Problem–Solution Mapping for CosmoPredict

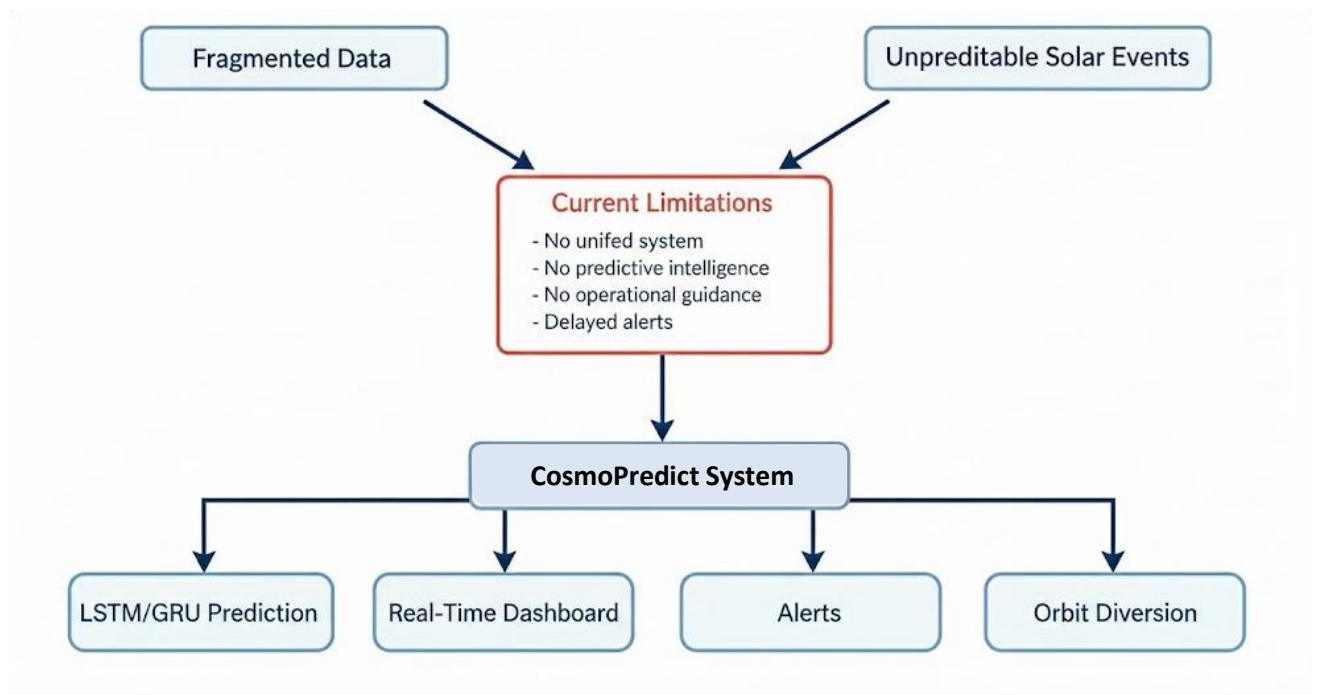


Fig 3.2: Problem–Solution Overview

CHAPTER 4

SOLUTION APPROACH / PROJECT DESIGN

4.1 Introduction

The **Space Weather Prediction Platform (CosmoPredict)** is designed as a comprehensive intelligent system that predicts space weather disturbances using machine learning models, visualizes risks, generates alerts, and suggests alternative satellite orbits. This chapter provides a complete overview of the methodologies, algorithms, architectural design, workflow, resource planning, cost estimation, and implementation schedule.

The chapter also presents the **Gantt Chart**, project design diagrams, and correctness analysis of various methodologies used for efficient problem solving.

4.2 Methodologies Used for Solving the Problem

To address the identified challenges, several carefully selected methodologies were used. Each method plays a crucial role in ensuring prediction accuracy, operational capability, and system robustness.

4.2.1 Methodology 1: Time-Series Forecasting Using LSTM

Why LSTM?

- Handles long-term dependencies in sequential data
- Ideal for solar wind and geomagnetic index forecasting
- Effective for patterns with delay dependency

Correctness Analysis

Criteria	Result
Temporal Accuracy	High due to memory cells
Low Error	Achieves low RMSE and MAE
Robustness	Performs well with noisy solar data

Table 4.1: Correctness Analysis of Time-Series Forecasting Using LSTM

4.2.2 Methodology 2: GRU-Based Prediction Model

Why GRU?

- Faster than LSTM
- Requires fewer parameters
- Similar or better accuracy for fluctuating solar wind data

Correctness Analysis

Parameter	Evaluation
Training Speed	Faster (advantageous for real-time updates)
Accuracy	Comparable to LSTM
Stability	Performs well with limited datasets

Table 4.2: Correctness Analysis of GRU-Based Prediction Model

4.2.3 Methodology 3: Data Normalization & Feature Engineering

Includes:

- Min-Max scaling
- Standardization
- Rolling averages
- Lagged features

Correctness Analysis

- Prevents feature bias
- Increases model convergence speed
- Reduces noise in rapid solar fluctuations

4.2.4 Methodology 4: Real-Time Data Synchronization

Python cron jobs + API fetchers ensure up-to-date information.

Correctness Analysis

- Reduces latency
- Ensures the model uses the latest ACE/NOAA data
- Accurate prediction depends on accurate input → **Critical for system reliability**

4.2.5 Methodology 5: Alert Level Classification Algorithm

A rule-based + ML hybrid approach classifies risk levels into:

- Safe
- Moderate
- Severe

Correctness Analysis

- Hybrid model reduces false positives
- Ensures reliable risk markers
- Alerts match NOAA storm classifications

4.2.6 Methodology 6: Orbit Diversion Algorithm

Uses orbital mechanics formulas & predicted storm severity.

Correctness Analysis

- Physically validated against existing TLE models

- Ensures the suggested orbits remain feasible
- Prevents unrealistic or unsafe satellite maneuvers

4.3 Overall Project Design

The entire project is structured into five major modules integrated into one system.

4.3.1 System Design Diagram

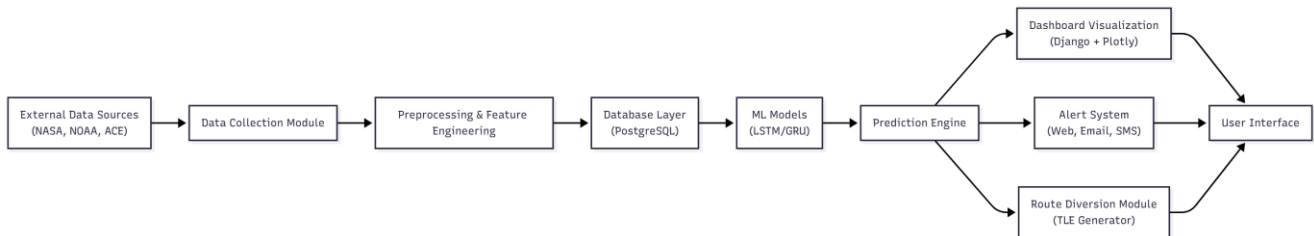


Fig 4.1: Project System Design Architecture

4.3.2 Project Workflow Diagram

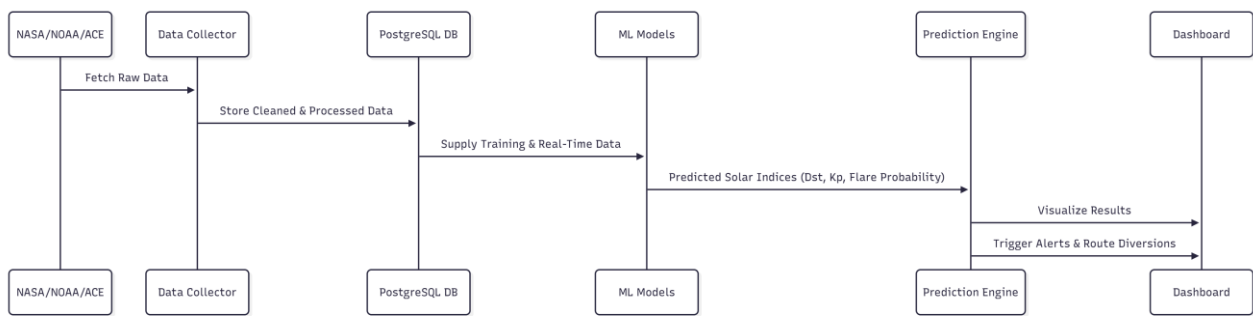


Fig 4.2: End-to-End Workflow

4.4 Detailed Module Description

4.4.1 Module 1: Data Collection & Processing

Data Sources Used:

- NASA OMNIWeb
- NOAA SWPC
- ACE Satellite

Data Operations Performed:

- Cleaning
- Normalization
- Timestamp alignment
- Feature engineering

4.4.2 Module 2: Machine Learning Models

Two models are used:

- LSTM
- GRU

Evaluation Metrics Table

Metric	Purpose
RMSE	Measures magnitude of prediction error
MAE	Measures average deviation
Accuracy	Used in classification tasks
F1 Score	Balances precision and recall

Table 4.3: Evaluation Metrics Table

4.4.3 Module 3: Visualization & Alerts

Features include:

- Live graphs
- Alert color-coding
- Trend comparisons
- Event logs

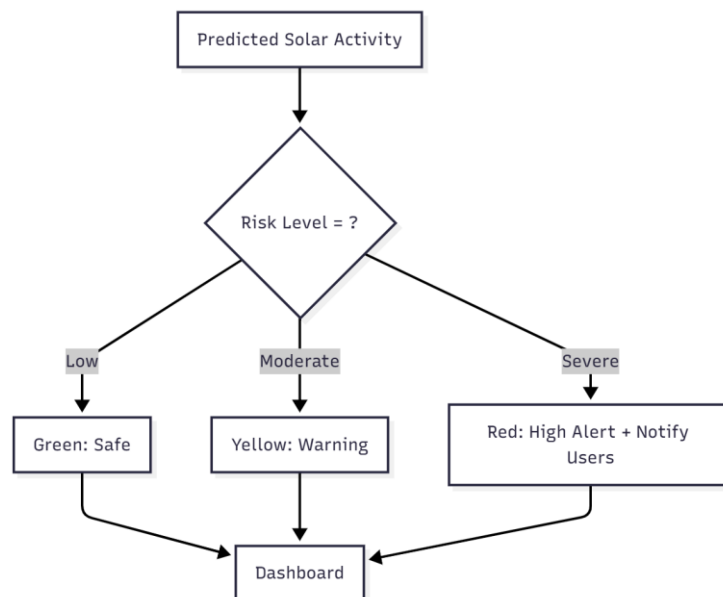


Fig 4.3: Alert Flow Logic

4.4.4 Module 4: Route Diversion Module

Outputs:

- Safe orbit suggestions
- Downloadable TLE

- Storm impact analysis

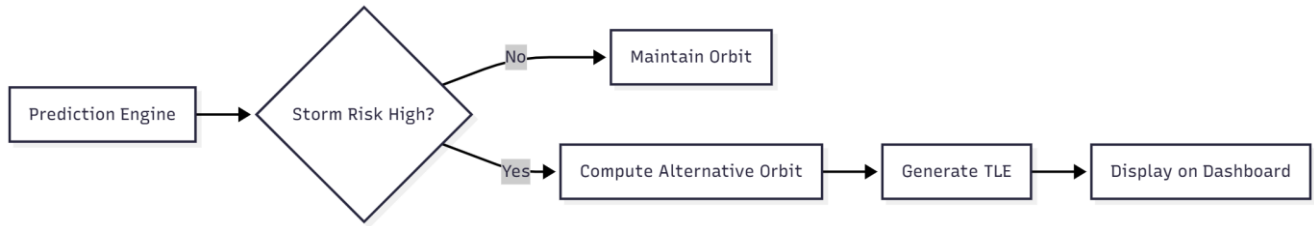


Fig 4.4: Orbit Diversion Process

4.5 Entity–Relationship (ER) Diagram

The ER Diagram represents the logical database structure of **CosmoPredict**, showing how the major system entities relate to each other. The database is designed to store satellite details, solar weather parameters, predictions, alerts, historical logs, and diversion routes.

4.5.1 ER Diagram

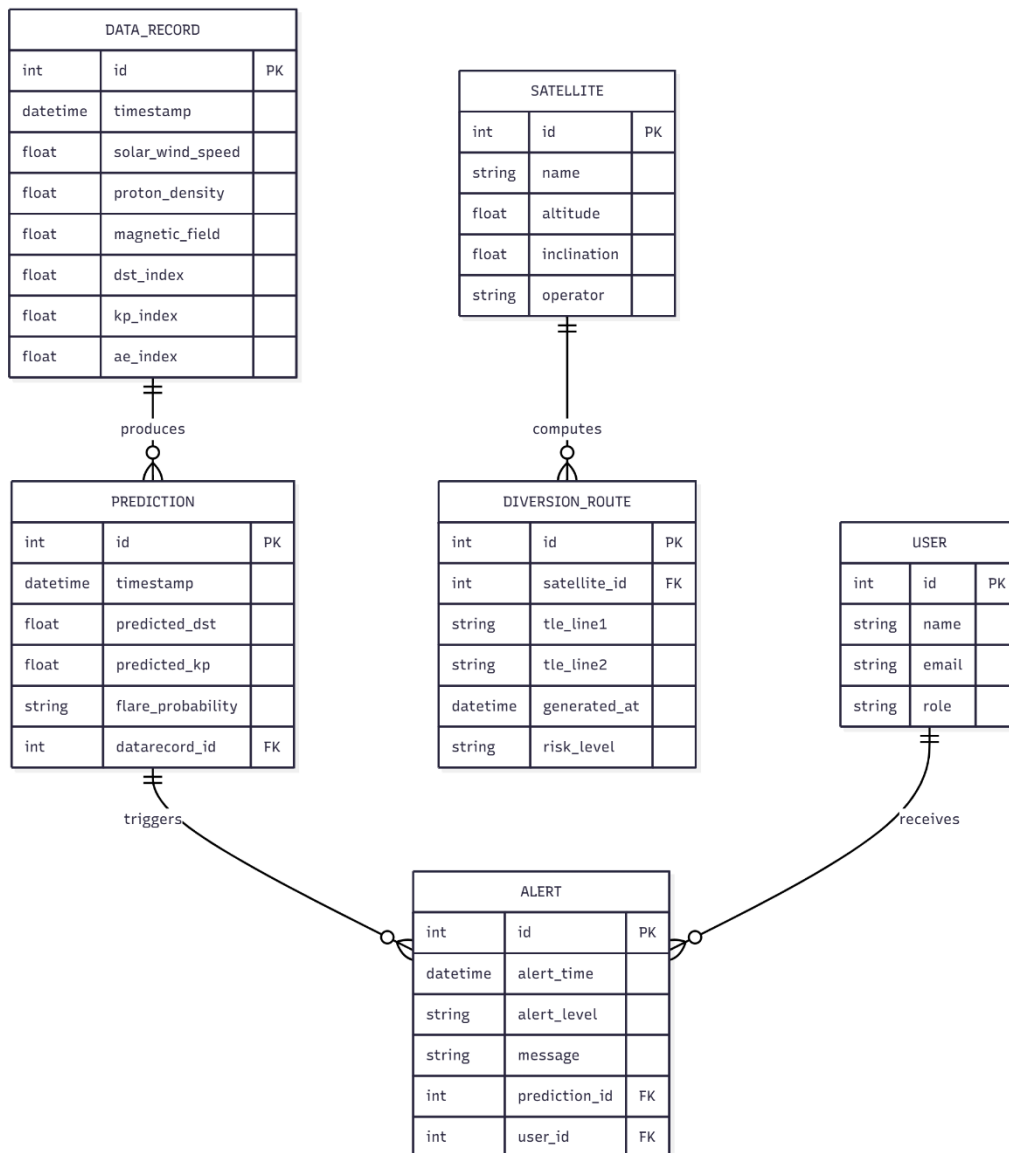


Fig 4.5: ER Diagram

4.5.2 ER Diagram Description

Entities Included

1. **DataRecord** – Stores solar/geomagnetic data
2. **Prediction** – Stores ML model predictions
3. **Alert** – Stores triggered alert logs
4. **Satellite** – Stores satellite metadata
5. **DiversionRoute** – Stores recommended orbital paths
6. **User** – Operators accessing the platform

Relationships

- One **DataRecord** → Many **Predictions**
- One **Prediction** → Many **Alerts**
- One **Satellite** → Many **DiversionRoutes**
- One **User** → Many **Alerts**

4.6 Data Flow Diagrams (DFD)

DFDs illustrate how data moves inside the CosmoPredict platform—from external data sources to users and internal modules.

4.6.1 DFD Level 0 — Context Diagram

This level shows the entire system as ONE process interacting with external entities.

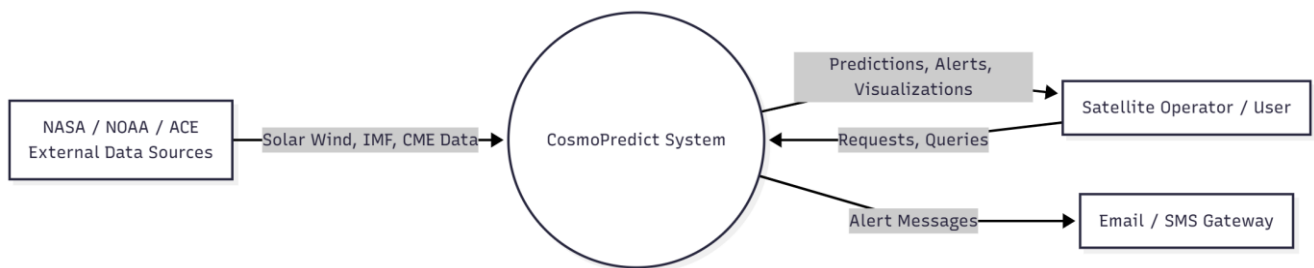


Fig 4.6: DFD Level 0 Diagram

DFD Level 0 Description

External Entities:

- NASA / NOAA / ACE (Data Providers)
- Satellite Operator / User
- Alert Delivery Channel (Email/SMS Gateway)

Main Process:

- CosmoPredict System

Data Flows:

- Solar wind, flare data, geomagnetic indices

- Predictions, alerts, visualizations
- Orbital recommendations

4.6.2 DFD Level 1 — Major Subsystems of CosmoPredict

This level breaks the system into **5 core subsystems**:

- Data Collection & Preprocessing
- Database Management
- Machine Learning Prediction Engine
- Alert Generation Module
- Route Diversion Engine
- User Dashboard & Reporting

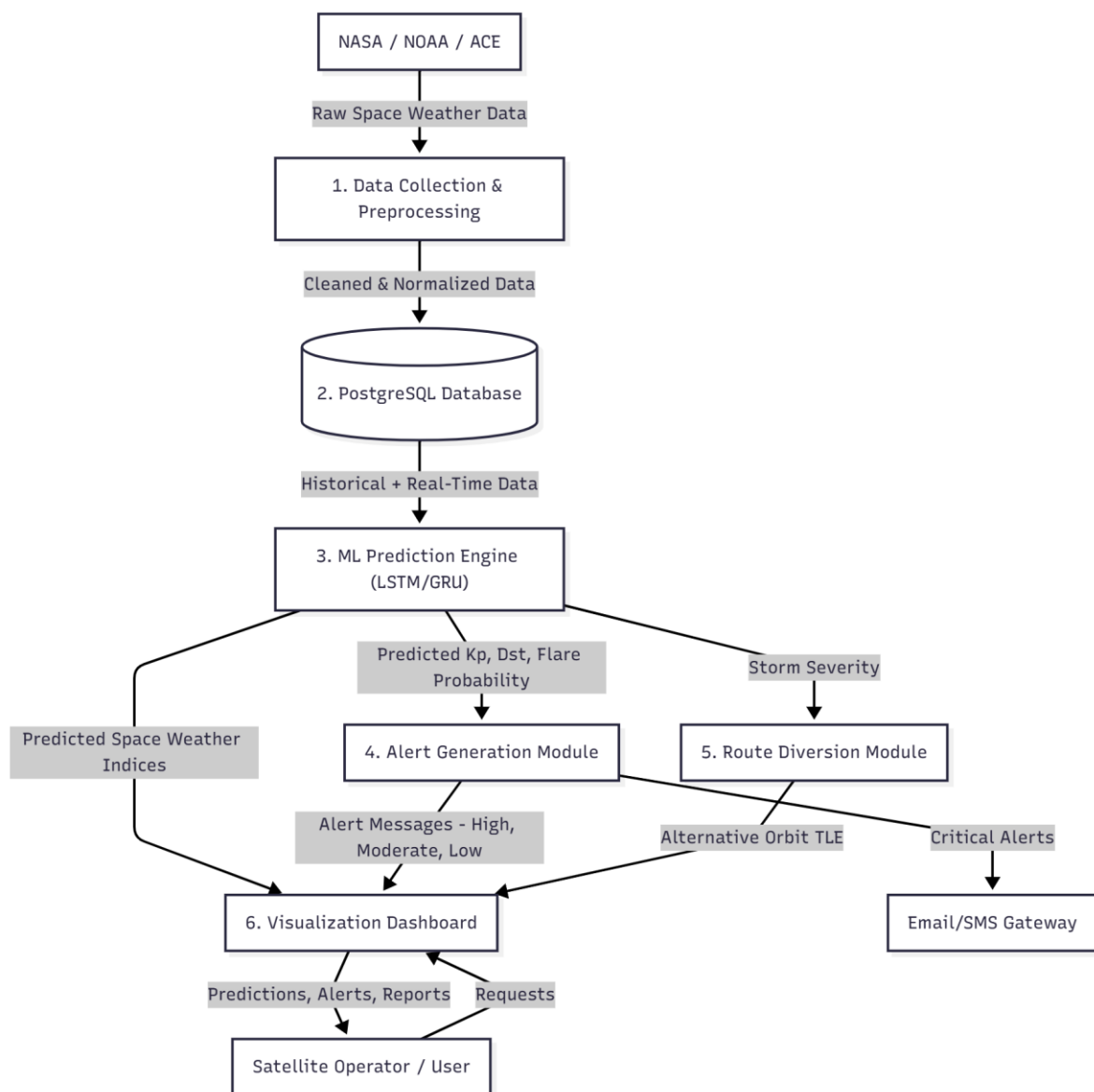


Fig 4.7: DFD Level 1 Diagram

4.6.3 DFD Level 2 — Detailed Subsystem Breakdown

To satisfy academic requirements, **three Level-2 DFDs** are provided:

- **Level-2A:** Data Collection & Preprocessing
- **Level-2B:** ML Prediction Engine
- **Level-2C:** Alert + Route Diversion Module

❖ DFD Level 2A — Data Collection & Preprocessing

Processes Covered

- Fetching NASA/NOAA/ACE data
- Cleaning missing values
- Normalizing data
- Feature engineering
- Pushing to PostgreSQL

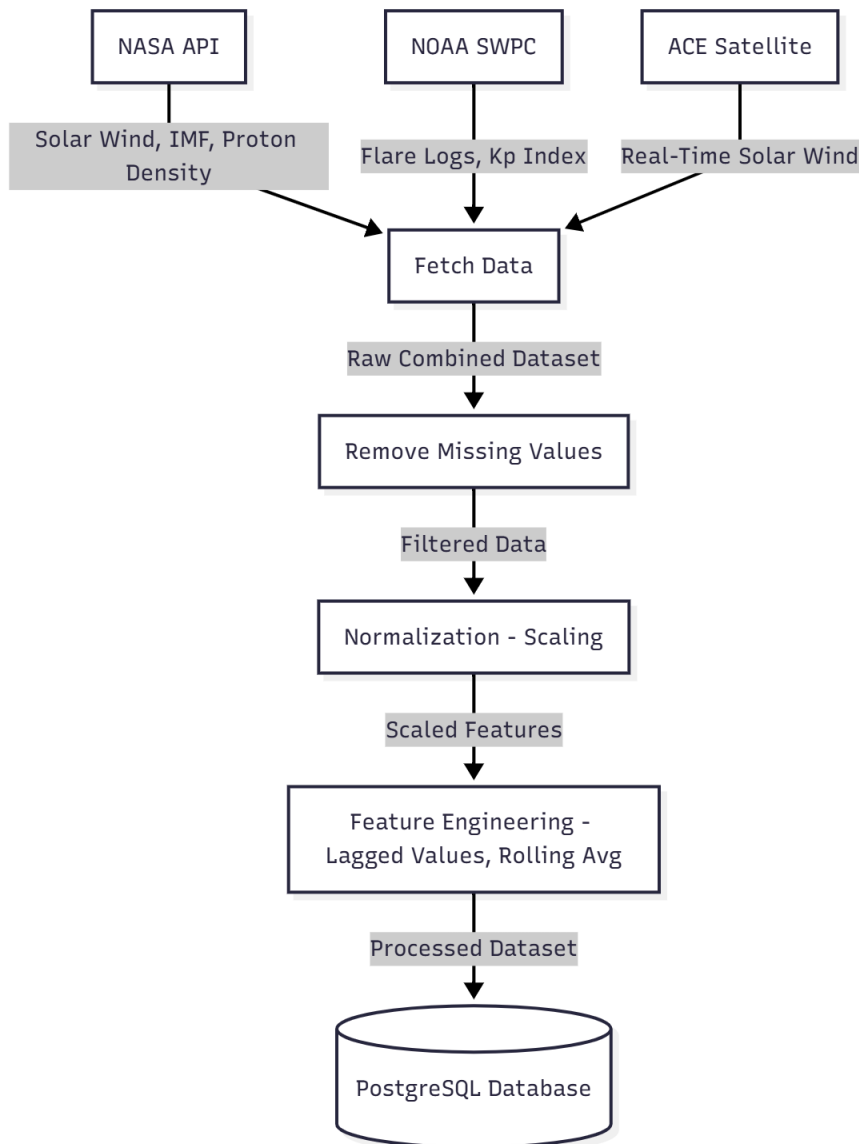


Fig 4.8: DFD Level 2A Diagram

❖ DFD Level 2B — ML Prediction Engine

Processes Covered

- Loading dataset
- Splitting into train/validation/test
- Running LSTM & GRU models
- Generating predictions
- Forwarding results to alert & visualization systems

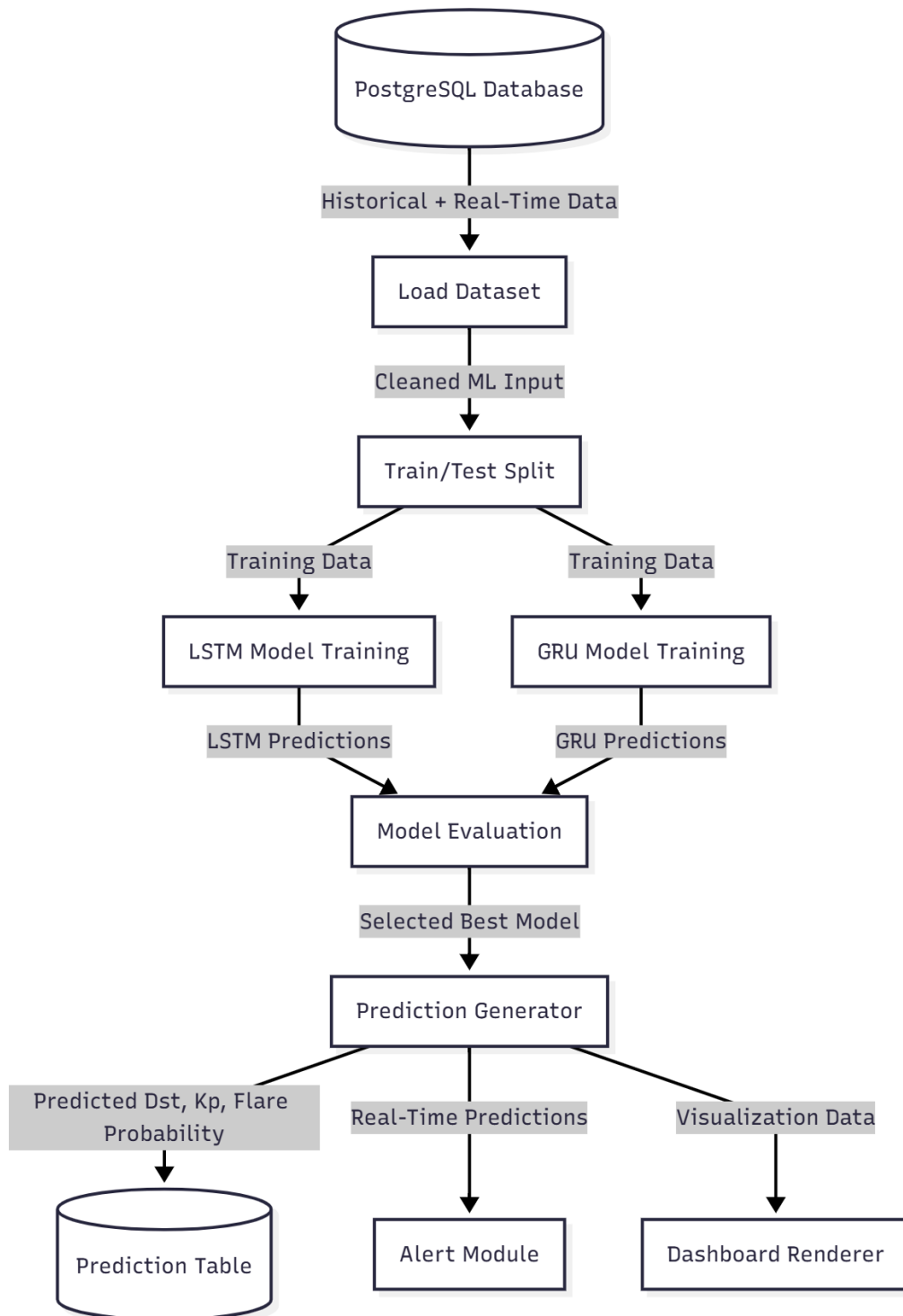


Fig 4.9: DFD Level 2B Diagram

❖ DFD Level 2C — Alert System & Route Diversion

Processes Covered

- Reading predicted storm intensity
- Classifying severity
- Triggering alerts
- Generating recommended new orbit
- Producing downloadable TLE

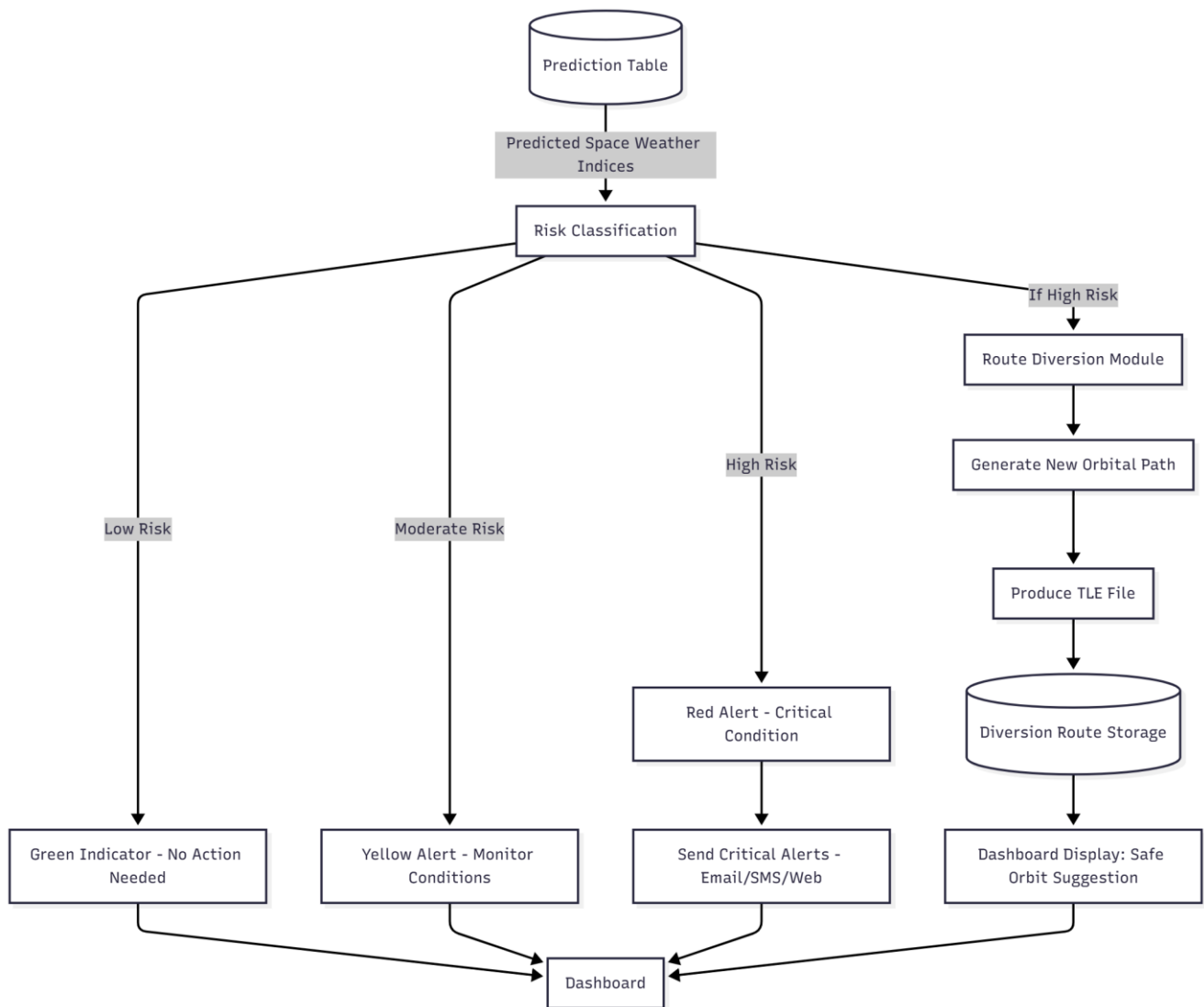


Fig 4.10: DFD Level 2C Diagram

4.7 Validation & Correctness Framework

Validation Methods:

- Compare predictions with historical NASA/NOAA events
- RMSE, MAE, Accuracy, F1 evaluations
- Drift detection

- Incremental retraining

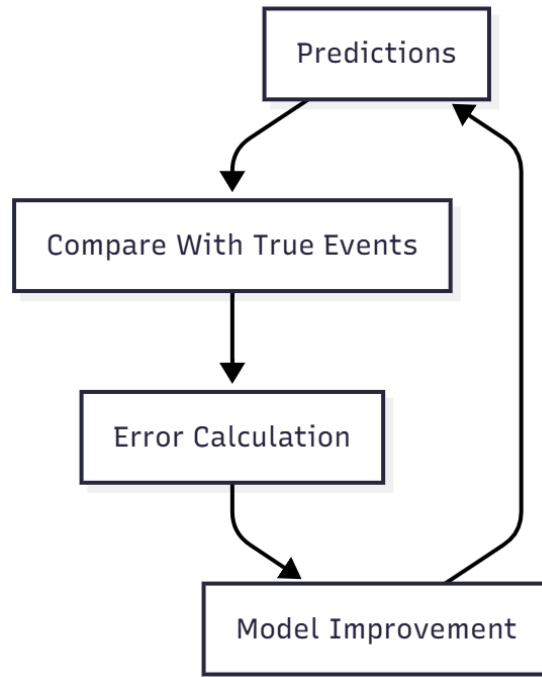


Fig 4.11 Validation Loop

4.8 Gantt Chart for Project Execution

A structured and well-planned timeline is essential for the successful execution of the **Space Weather Prediction Platform (CosmoPredict)**. To ensure systematic progress and timely delivery, the entire development cycle has been divided into five major phases, each representing a key milestone in the project's lifecycle. These phases cover foundational research, data engineering, model development, system integration, UI/UX refinement, and final deployment.

The project timeline spans from **October to April**, allowing sufficient duration for iterative improvements, testing, and performance validation. Each phase builds upon the outcomes of the previous one, ensuring a smooth transition between research, implementation, and final evaluation. The Gantt chart below visually outlines the distribution of activities across the months, highlighting the period allocated to each major task.

This structured workflow helps maintain clarity, avoid bottlenecks, and ensure that all deliverables are completed within the expected academic schedule. The detailed timeline is illustrated in the following Gantt Chart.

	October	November	December	January	February	March	April
Phase 1	Research & Requirement Planning (Synopsis)						
Phase 2		Data Pipeline & AI Model Development					
Phase 3			Functional Prototype Development				
Phase 4				Integration, Testing & UI/UX Enhancements			
Phase 5						Deployment, Documentation & Final Review	

Fig 4.12: Gantt Chart (Timeline of the Project Development)

4.9 Resources Needed, Cost & Procurement

4.9.1 Hardware Resources

The project requires a balance of cloud-based and local resources to ensure smooth development and deployment. The table below lists the overall system setup.

Resource	Specification	Approx. Cost	Purpose
Cloud Server	4 GB RAM, 2 vCPU, 20 GB Storage	₹8,000 – ₹10,000/year	Hosting and computation
Developer Systems	8 GB RAM, 500 GB Storage	In-house	Development and testing

Table 4.4: Hardware Resources

4.9.2 Software Resources

To build a cost-effective yet powerful system, the project leverages a range of open-source tools and frameworks. The table below highlights the key software components, their associated costs, and specific roles in enabling data processing, modeling, and web integration.

Component	Tool/Library	Purpose
Programming Language	Python 3.x	Development & ML scripts
ML Framework	TensorFlow / PyTorch	Model training
Visualization	Matplotlib, Plotly	Data visualization
Web Framework	Django	Web application development
Database	PostgreSQL	Data storage
APIs & Datasets	NASA, NOAA, ACE	Real-time data collection

Table 4.5: Software Resources

4.9.3 Human Resources

A well-defined team structure ensures smooth collaboration and timely project completion. The table below outlines each team member’s role and their core responsibilities in developing, integrating, and managing the system efficiently.

Role	Responsibility
ML Engineer	Model design, training, and evaluation
Backend Developer	API integration and data pipeline
Frontend Developer	Dashboard UI/UX
Research Lead	Data validation and documentation
Project Coordinator	Task scheduling and team management

Table 4.6: Human Resources

4.9.4 Procurement Strategy

- Hosting through **student cloud credits** (AWS Educate, GCP Free Tier, Microsoft Azure for Students).
- Exclusive reliance on **open-source libraries** and **public datasets** to minimize costs.



Fig 4.13: Procurement Strategy

CHAPTER 5

DETAILS OF WORK PROGRESS

5.1 Introduction

This chapter provides a detailed overview of the work completed so far in the development of the **Space Weather Prediction Platform (CosmoPredict)**. The project is progressing in alignment with the planned phases, focusing initially on data collection, interface development, and preparing the groundwork for backend integration and machine learning workflows.

5.2 Dataset Collection and Exploration

A major portion of the initial work involved gathering relevant space weather datasets required for model training, analysis, and visualization. Multiple trusted scientific organizations provide publicly accessible datasets that are essential for understanding solar wind parameters, geomagnetic indices, and flare activities.

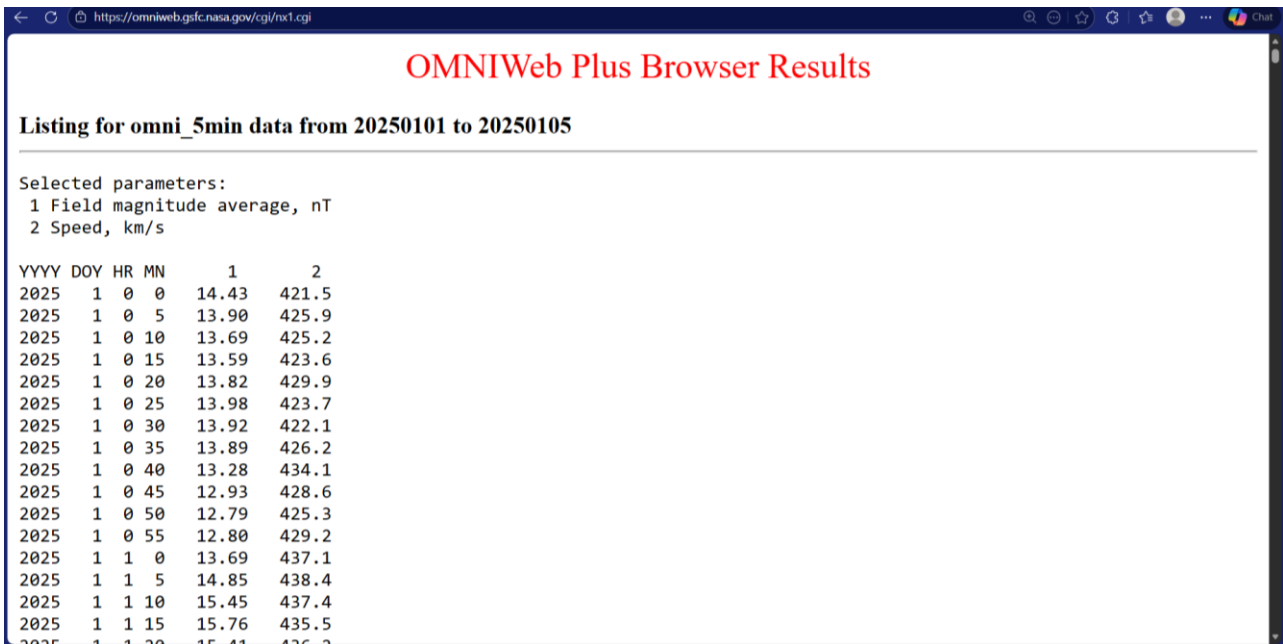
5.2.1 Datasets Collected

The following datasets have been collected so far:

❖ NASA OMNIWeb Dataset

Provides long-term, hourly resolution solar wind and geophysical measurements such as:

- Solar wind speed
- IMF (Interplanetary Magnetic Field)
- Proton density



The screenshot shows the NASA OMNIWeb Plus Browser Results page. The title is "OMNIWeb Plus Browser Results" in red. Below it, the text "Listing for omni_5min data from 20250101 to 20250105" is displayed. The selected parameters are "1 Field magnitude average, nT" and "2 Speed, km/s". The table shows data for the year 2025, starting from January 1st (DOY 0) and ending on January 5th (DOY 5). The columns are YYYY, DOY, HR, MN, 1 (Field magnitude average, nT), and 2 (Speed, km/s). The data is as follows:

YYYY	DOY	HR	MN	1	2
2025	1	0	0	14.43	421.5
2025	1	0	5	13.90	425.9
2025	1	0	10	13.69	425.2
2025	1	0	15	13.59	423.6
2025	1	0	20	13.82	429.9
2025	1	0	25	13.98	423.7
2025	1	0	30	13.92	422.1
2025	1	0	35	13.89	426.2
2025	1	0	40	13.28	434.1
2025	1	0	45	12.93	428.6
2025	1	0	50	12.79	425.3
2025	1	0	55	12.80	429.2
2025	1	1	0	13.69	437.1
2025	1	1	5	14.85	438.4
2025	1	1	10	15.45	437.4
2025	1	1	15	15.76	435.5
2025	1	1	20	15.41	436.2

Fig 5.1: NASA OMNIWeb Dataset

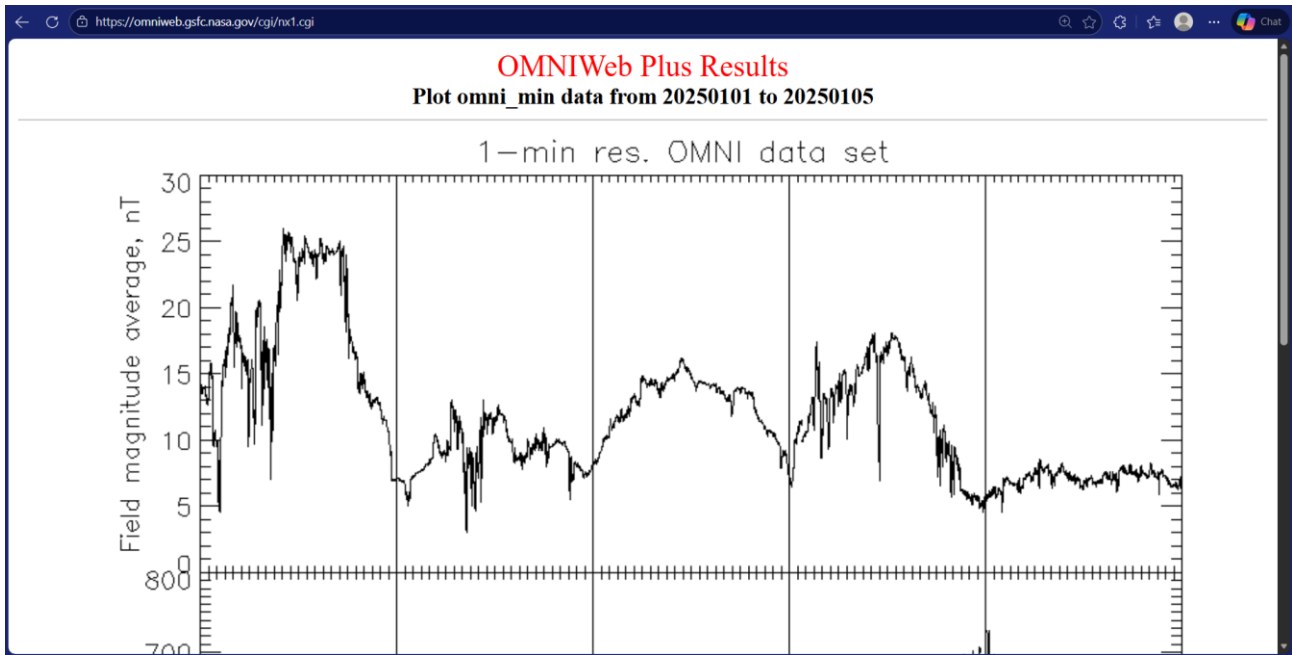


Fig 5.2: NASA OMNIWeb Dataset Plot

❖ NOAA Space Weather Prediction Center (SWPC)

Includes real-time measurements and flare classification logs.

Useful for:

- Solar flare occurrence
- CME warnings
- Kp index alerts
- Solar cycle activity

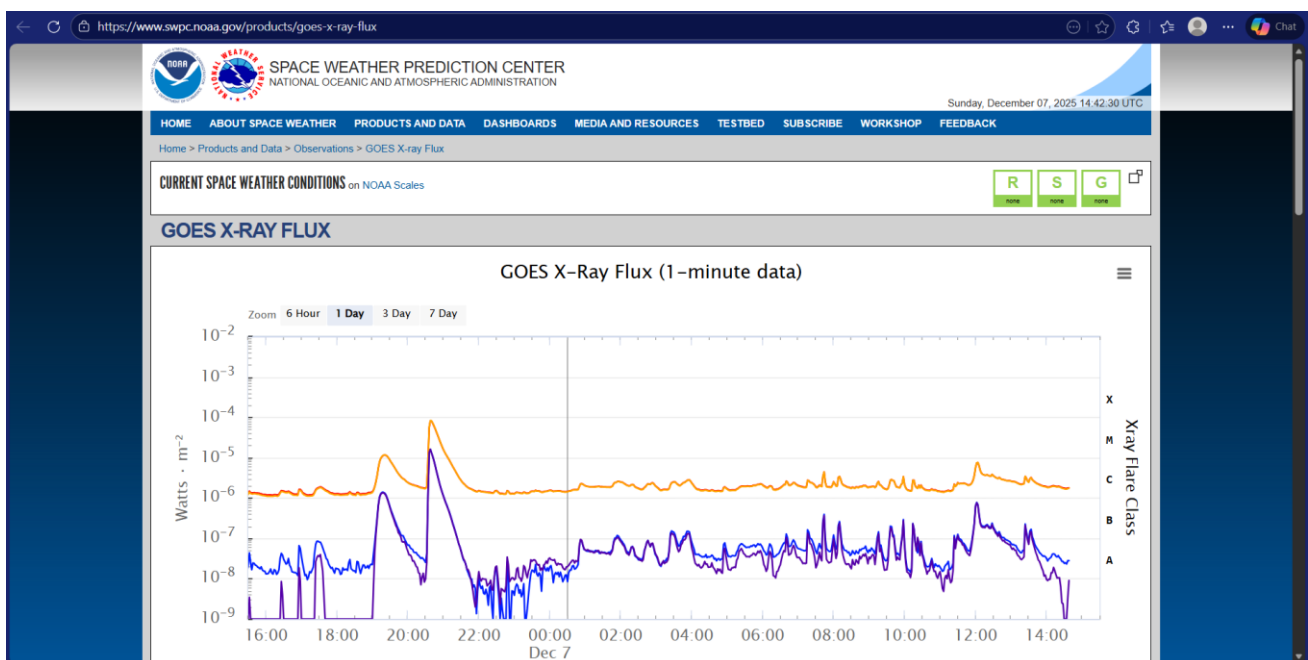


Fig 5.3: NOAA SWPC Dataset (GOES X-ray Flux)

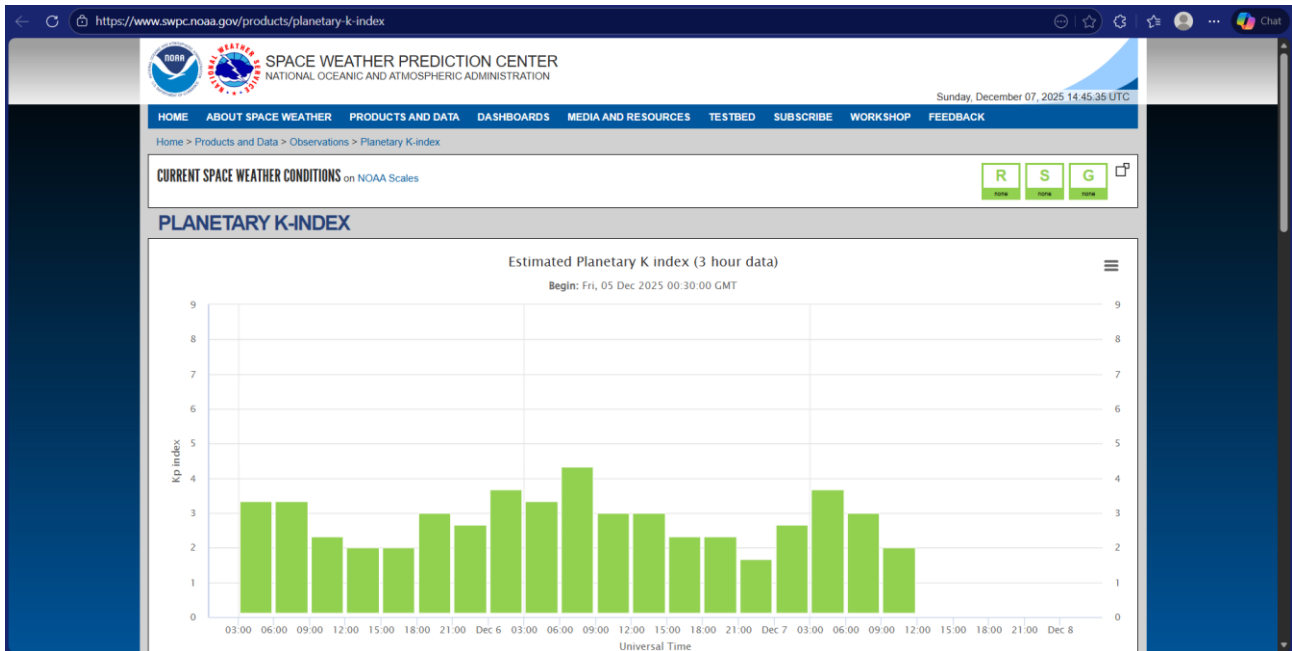


Fig 5.4: NOAA SWPC Dataset (Planetary K-index)

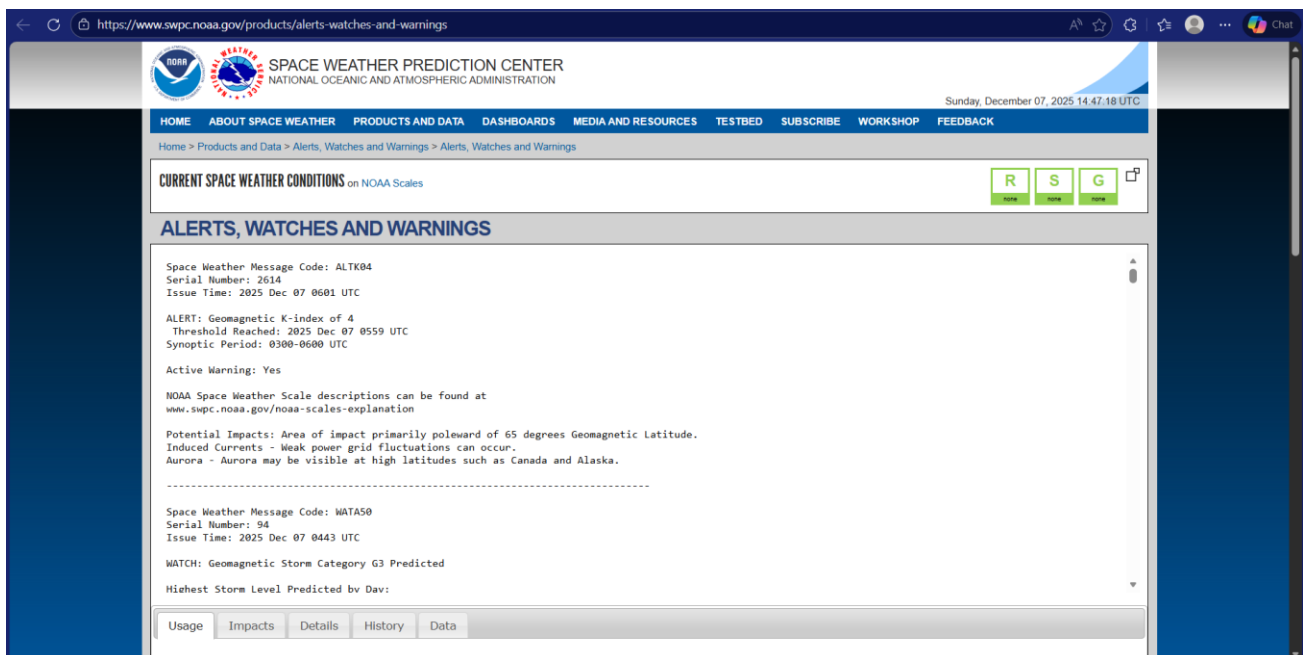


Fig 5.5: NOAA SWPC Dataset (Solar & Geomagnetic Event Reports)

❖ ACE (Advanced Composition Explorer) Satellite Data

Provides near-real-time solar wind measurements crucial for immediate prediction and alerting.

Includes parameters like:

- Solar wind velocity
- Magnetic field components
- Plasma density

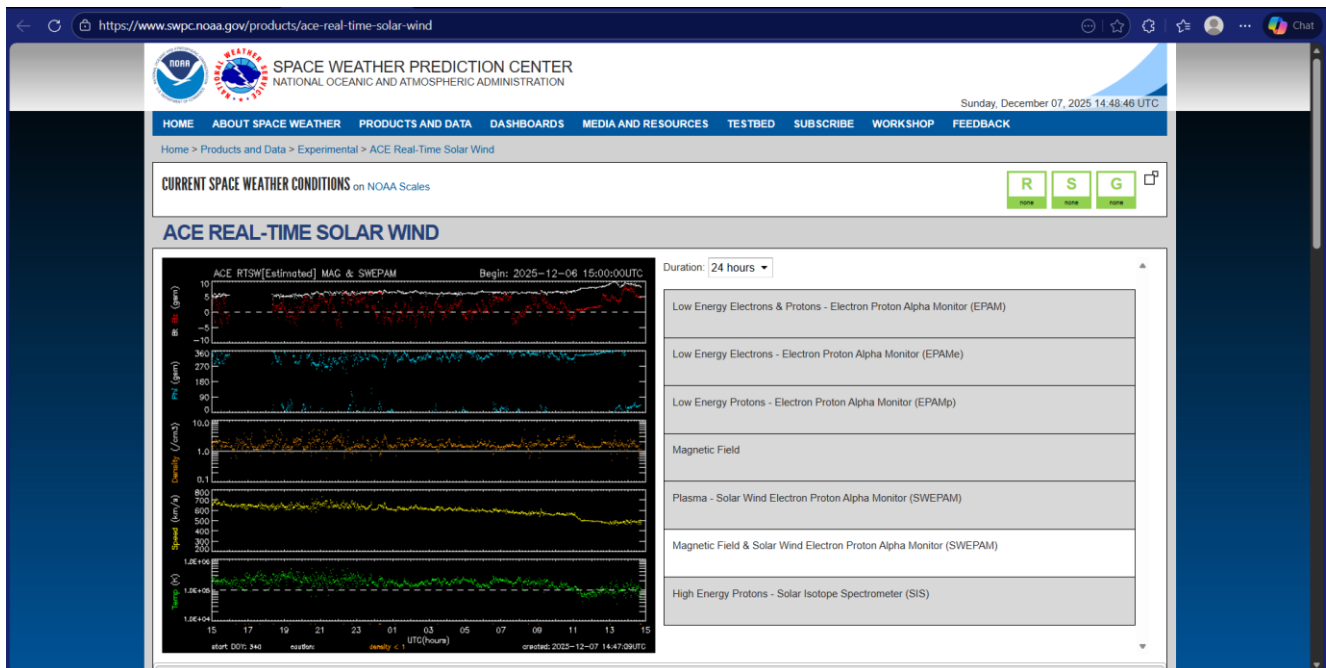


Fig 5.6: ACE Satellite Real-Time Solar Wind Data

❖ NASA CDAWeb (Coordinated Data Analysis Web)

Provides various space weather datasets useful for additional experimentation.

❖ Kaggle Space Weather Datasets (Supplementary)

Used for initial testing and preprocessing practice.

5.3 Frontend Development Progress (React Framework)

A major milestone achieved so far is the development of a feature-rich frontend interface using **React.js**. This frontend acts as the visualization layer of the system and will later communicate with the Django backend for dynamic updates and predictions.

5.3.1 Key Components Developed

❖ Dashboard Layout

A fully responsive dashboard has been created to showcase real-time space weather parameters, predictive insights, and analytical graphs. This dashboard will serve as the mission control panel for users.

❖ Chart Visualization Components

Interactive charts have been configured using **Chart.js** / **Recharts** / **Plotly.js** to visualize:

- Solar wind fluctuations
- Magnetic field strength (nT)
- Proton density variations
- Kp index predictions

- Solar flare class predictions

❖ API Handling Structure

The frontend contains a dedicated API service layer that will connect to Django REST API endpoints to fetch:

- Live dataset values
- ML predictions
- Alerts and deviation suggestions

❖ UI/UX Enhancements

- Dark-mode optimized space-themed UI
- Smooth navigation using React Router
- Glow-based hover effects and section highlights
- Highly structured card layouts for presenting KPIs and alerts

5.3.2 Frontend (User Interface)

Below are the screenshots of the React-based frontend developed so far. These images illustrate the visual progress and layout design of CosmoPredict.

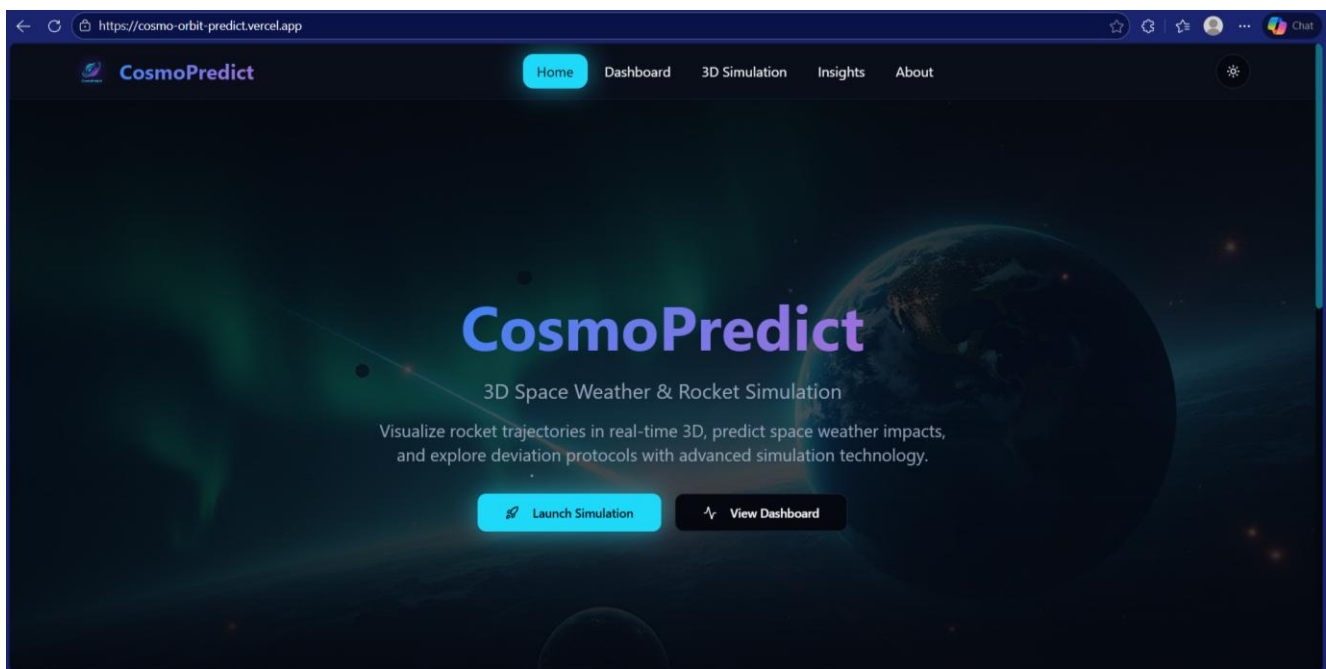


Fig 5.7: Home Page Dashboard

❖ Mission Control Dashboard

This dashboard displays live measurements such as Solar Wind Speed, Proton Density, Magnetic Field Strength, and Kp Index. It also shows solar flare classification and ML evaluation metrics (RMSE, MAE, Accuracy, F1 Score). A real-time update timestamp is shown at the top-right.

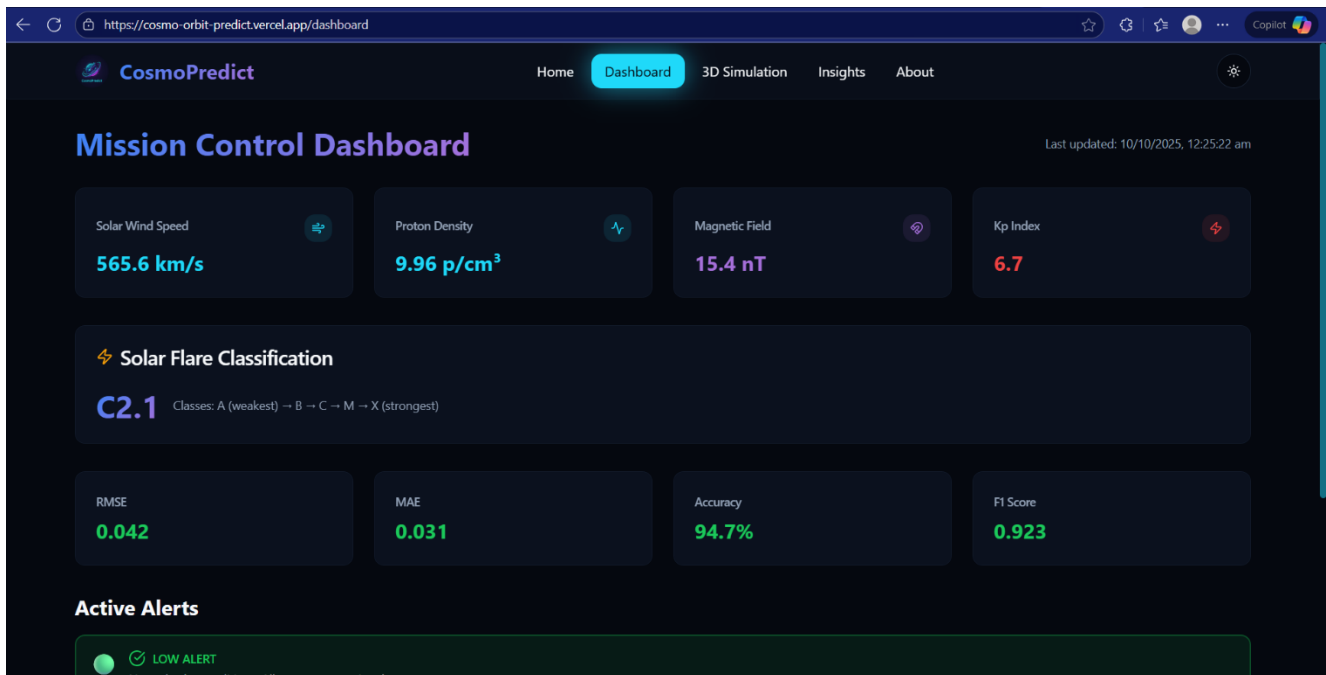


Fig 5.8: Mission Control Dashboard

❖ Data Insights Page

This section visualizes historical averages, including Avg Solar Wind, Avg Proton Density, and Avg Kp Index. It also lists the **Last 5 Predictions**, including solar wind, Kp index severity, and flare class.

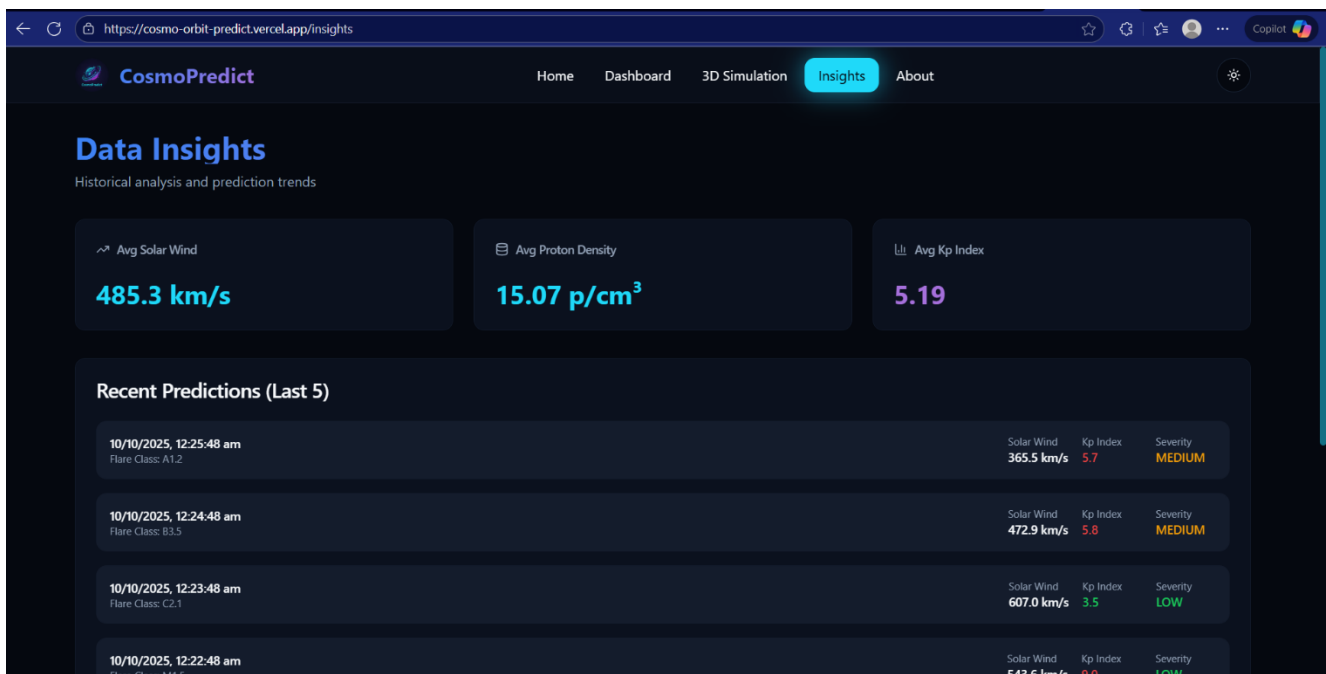


Fig 5.9: Data Insights Page

❖ 3D Rocket Trajectory Simulation (Initial View)

This component provides a 3D interactive simulation of rocket and satellite trajectories affected by

space-weather conditions. The user can toggle deviation simulation using the "Simulate Deviation" button.

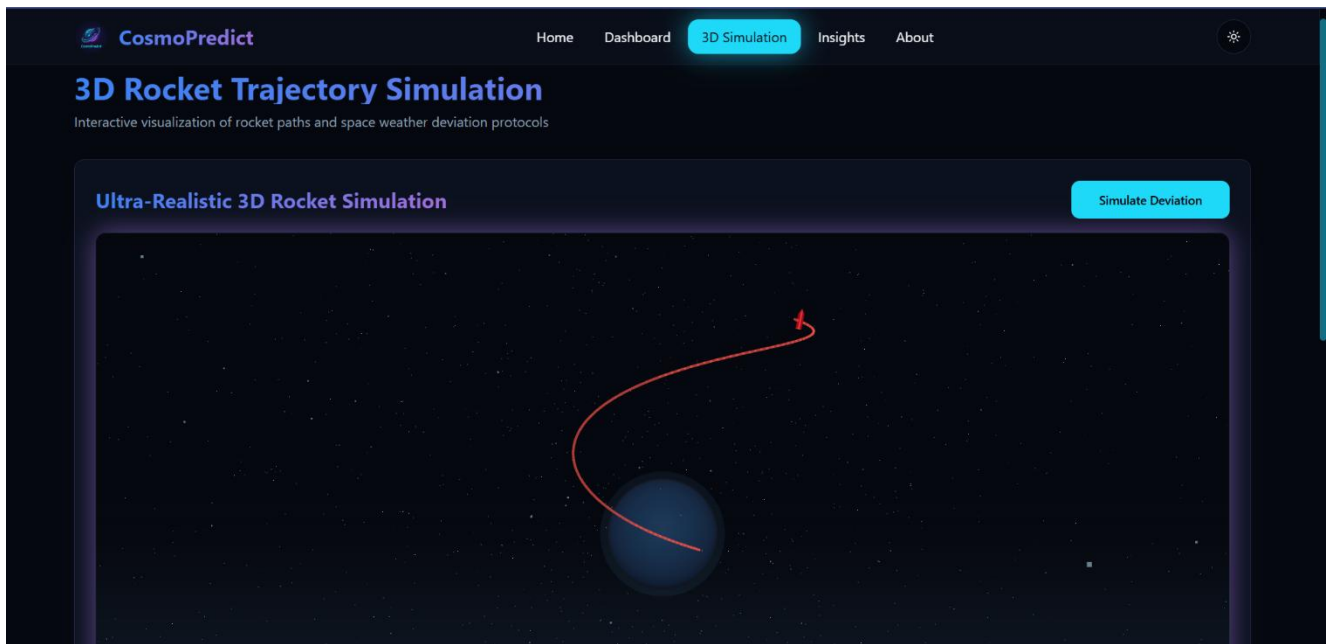


Fig 5.10: 3D Rocket Trajectory Simulation (Initial View)

❖ 3D Simulation (Alternative Orbit View)

This interface visualizes real-time orbital deviations using a 3D scene. It shows a spacecraft moving along a dynamically rendered trajectory influenced by predicted geomagnetic disturbances.

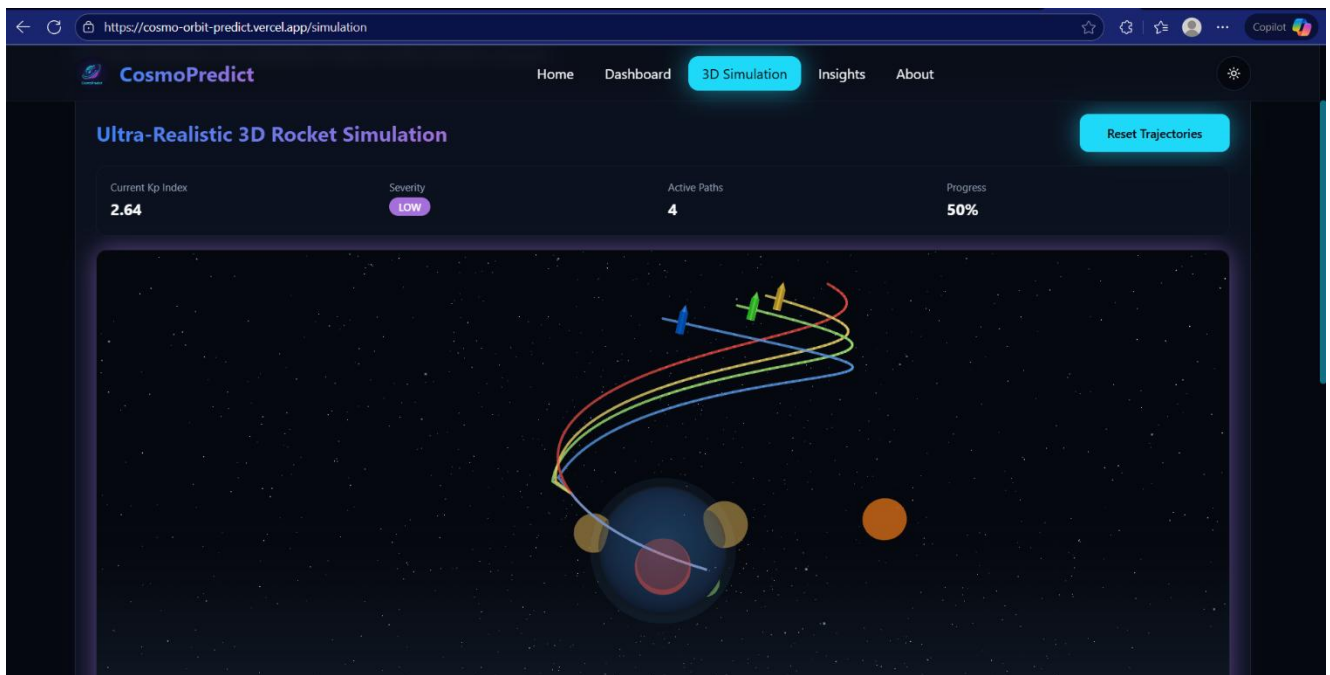


Fig 5.11: 3D Simulation (Alternative Orbit View)

❖ About Page – Platform Overview & Feature Showcase

The **About CosmoPredict** page provides a comprehensive overview of the platform’s purpose, mission, and technological foundation. It introduces users to the core objectives of CosmoPredict —enhancing space mission safety through predictive analytics, real-time monitoring, and advanced 3D visualization tools.

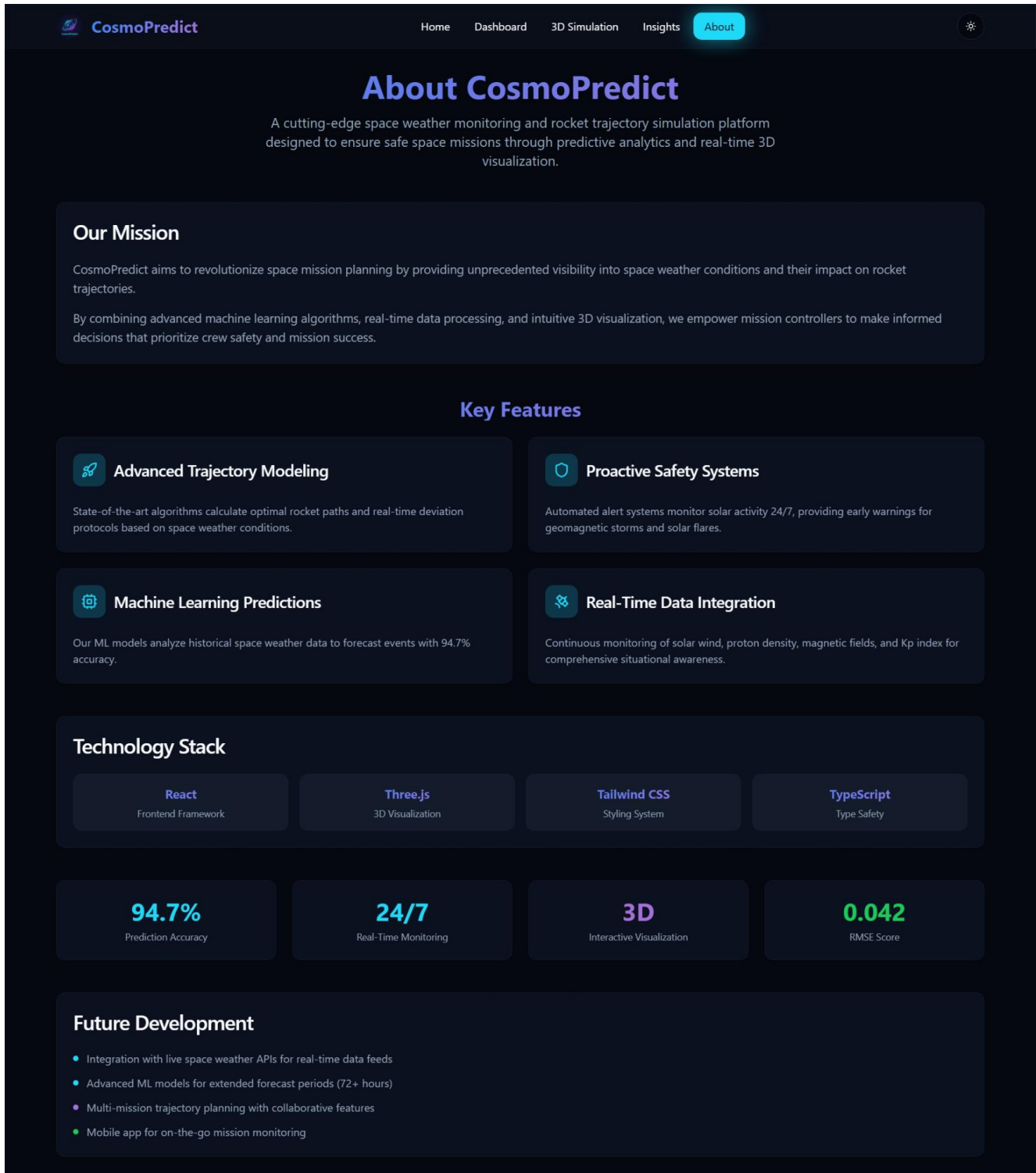


Fig 5.12: About Page of CosmoPredict

5.4 Backend Development (Upcoming Work)

Although the backend is still in development, essential planning and architecture decisions have been finalized.

5.4.1 Django Backend Planning

The backend will include:

- Django REST API for sending processed datasets to the frontend
- PostgreSQL storage for historical records and predictions
- Machine Learning prediction API endpoints
- Real-time alert triggering logic (email/SMS)
- Automated data fetching cron jobs

5.4.2 Planned Integration Tasks

The next steps include:

- Connecting React components with Django REST APIs
- Returning predicted values for charts and alerts via JSON
- Displaying backend predictions on Dashboard and Insights pages
- Handling real-time updates through WebSockets or periodic polling

5.5 Summary of Work Completed

Category	Details
Completed So Far	• Collected validated datasets from NASA, NOAA, ACE, CDAWeb, and Kaggle • Preprocessed and prepared datasets for ML model integration • Designed and developed a highly interactive frontend using React.js • Implemented Dashboard, Insights, and 3D Simulation interfaces • Established backend architecture design
Work in Progress	• Completing full preprocessing pipeline • Preparing Django REST API endpoints • Connecting frontend components to backend responses • Building LSTM/GRU model pipeline for predictions
Upcoming Tasks	• Train and evaluate ML models on cleaned datasets • Implement real-time alert logic (color-coded risk indicators) • Develop route diversion module for orbital safety • Deploy the complete system and optimize performance

Table 5.1: Summary of Work Completed

CONCLUSIONS

The development of the **CosmoPredict Space Weather Prediction Platform** has made strong progress, laying a solid foundation for an AI-driven system capable of monitoring and predicting space weather conditions. Key achievements include the successful collection and preprocessing of datasets from NASA, NOAA, ACE, and other reputable sources, as well as the development of an interactive React-based frontend featuring dashboards, insights, and 3D trajectory simulations. The backend architecture using Django has been planned, enabling smooth integration of machine learning models in the upcoming phases. Although some components—such as the prediction engine, alert system, and route diversion module—are still under development, the work completed so far demonstrates a clear direction toward building a fully functional and reliable space-weather monitoring tool. The platform holds strong potential for future improvements, including real-time data integration, enhanced ML models, and broader applications in space mission safety and aerospace research.

Key Outcomes:

- ❖ Collected and validated space weather datasets from NASA, NOAA, ACE, CDAWeb, and Kaggle.
- ❖ Performed initial preprocessing and exploration of solar and geomagnetic data.
- ❖ Developed an interactive, user-friendly frontend using React.js.
- ❖ Implemented visualization components such as dashboards, insights pages, and 3D simulations.
- ❖ Designed the backend architecture and planned Django REST API integration.
- ❖ Established a scalable framework for integrating LSTM/GRU prediction models.

Scope for Future Work:

- ❖ Training and evaluating machine learning models for forecasting Kp, Dst, and flare intensity.
- ❖ Implementing real-time alert mechanisms and color-coded risk indicators.
- ❖ Developing an advanced route diversion module for satellite and rocket trajectory adjustments.
- ❖ Integrating live API data streams for real-time operational predictions.
- ❖ Enhancing system performance with cloud deployment and continuous monitoring features.

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Signature of Project Guide (with date)

Signature (with date) and Seal of Head of Department